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Magnetic Levitation Control

Automation and Control Laboratory

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1 Introduction

In this laboratory project, a real-time magnetic levitation setup (MagLev) was used to explore various control techniques aimed at regulating the current through the electromagnet and stabilizing or tracking the vertical position of the steel ball. Magnetic levitation system is a nonlinear and inherently unstable platform consists of:

- an **electromagnet**, composed of a solenoid and an iron core, which, when powered by a voltage, generates a magnetic force;
- a **ball made of ferromagnetic material**;
- an **optical sensor** that measures the **vertical position** of the ball;
- a **current sense resistor** in series with the solenoid, enabling the measurement of the **current** $i(t)$;
- a controllable input: the **applied voltage** $u(t)$ to the solenoid.



Figure 1: Image of the system

Note: The main constraint on the control input arises from the voltage limitations of the system, which can accept a maximum of 24V. Since the amplifier applies a gain of 3 to the signal received from the DAQ, the maximum permissible DAQ output voltage is approximately 8V. According to the datasheet, the DAQ can provide up to 10V, which in turn corresponds to a maximum amplified output of 30V. However, to ensure safe operation and reduce the risk of entering nonlinear regions, the control input is conservatively limited to 23V. [2]

Considering the characteristics of the system under analysis, the following control challenges have been identified and addressed:

- **Task 1: Electromagnetic Coil Current Regulation (Frequency-Based Control)**

The aim of this task is to implement a frequency-domain control strategy to regulate the current flowing through the electromagnetic circuit. The primary focus is on ensuring a stable and accurate response of the coil current under varying operating conditions.

- **Task 2: Ball Position Stabilization (Frequency-Based + State-Space Control)**

This task is concerned with maintaining the vertical position of the ferromagnetic ball at a desired reference point. A combination of frequency-domain and state-space control methodologies is employed to achieve robust stabilization in the presence of system nonlinearities and external disturbances.

- **Task 3: One-Dimensional Ball Trajectory Tracking and Lift-Off (State-Space + Adaptive Control)**

The objective here is to design a control framework that enables the ball to lift off from its resting point and follow a predefined vertical trajectory. This is accomplished through a synergy of state-space control techniques and adaptive control strategies, allowing the system to respond effectively to parameter variations and uncertainties during the motion.

2 Model

2.1 Sensors

The system is composed of two different sensors that will be used to measure: the vertical position of the steel ball and the current in the windings.

2.1.1 Position Sensor

The sensor consists of an NPN silicon photodarlington, which is a light-sensitive device that is used to measure the vertical position of the steel ball. The output of the sensor is a voltage that can be converted in position using the following formula.

$$\hat{z} = k_b(V + q_{PS})$$

- $\hat{z}$ is the vertical position of the ball
- k_b is the sensor sensitivity that converts voltage to meters
- V is the measured voltage
- q_{PS} is the voltage offset

Sensor calibration is required to calculate parameters that are not provided in the datasheet or whose values may have deviated from those specified in the datasheet [1].

The procedure for estimating the voltage offset is described below:

- The ball was placed on the post in the chamber.
- A series of tests were conducted using step inputs of varying amplitudes, selected to ensure that the ball remained in its initial position.
- Since the ball remained stationary throughout the experiments, the measured position is expected to be zero for the entire duration of each test. However, the presence of a voltage offset causes the measurements to deviate from zero.
- This offset is estimated by computing the average of the recorded position values. Specifically, the estimated offset is $q_{PS} = -0.0231V$.

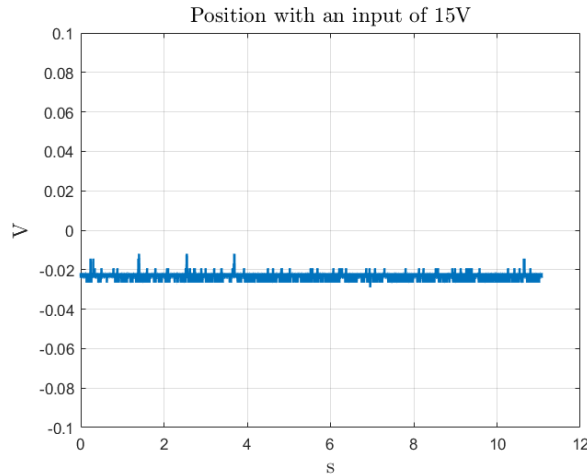


Figure 2: Sensor Position Offset

As for the sensor sensitivity, the following approach has been adopted:

- Step input tests were performed in such a way that the ball would reach a known vertical position. The reference position was defined as the point at which the ball reached the upper limit (in tests where the applied voltage exceeded 21.3V) after the transient response had settled. The outputs corresponding to these tests should therefore reflect a vertical position of 12mm, which is the maximum distance of the air gap.
- To improve accuracy, an average was computed from the steady-state values, i.e., after the transient phase had decayed.
- For each test, the sensor sensitivity was calculated as the ratio between the known vertical position and the difference between the corresponding average steady-state output and the previously estimated voltage offset.
- A single representative sensitivity value was then obtained by averaging the results of all tests, obtaining $k_b = 0.0033 \text{ m/V}$.

2.1.2 Current Sensor

The current is measured using a current sense resistor with a resistance of 1Ω , placed in series with the electrical circuit. Since the output of the sensor is a voltage signal, the current can be calculated directly by dividing the measured voltage by the resistance of the sensor.

Just like the position sensor, the current sensor may exhibit an offset, meaning that it may output a non-zero current even when the actual input is zero.

To identify this offset, the following procedure was carried out:

- Step voltage tests of varying amplitudes were applied, ranging from 0V to 23V.
- For each test, only the data recorded before the voltage step (i.e., when the input voltage was zero) were considered.
- The remaining data were converted from voltage to current by dividing by the sensor resistance and then averaged to obtain a single current value for each test.
- The final offset value was computed by averaging the individual values obtained in the previous step across all tests.
- The identified offset value is $q_{CS} = -0.0166A$.

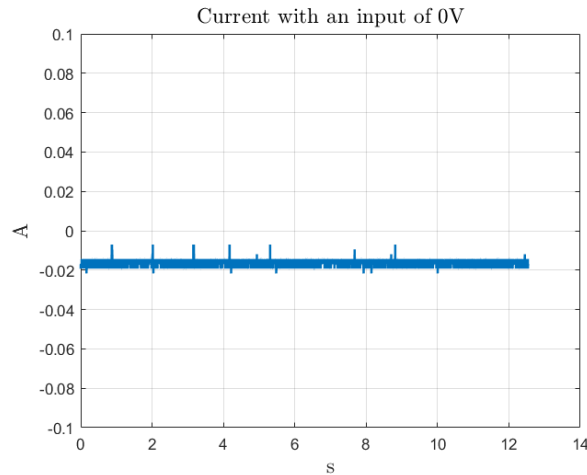


Figure 3: Current Sensor Offset

2.2 System Modeling

The following sections describe the steps taken to model the system and identify the state variables.

2.2.1 Force Balance and System Dynamics

To model the system's behavior, the analysis begins with the consideration of the forces acting on the ball. This approach is essential as it accounts for both external and internal influences on the ball's motion. The gravitational force acts downward, while the magnetic force—dependent on the solenoid current and the distance between the ball and the solenoid—acts upward. By applying Newton's second law of motion, the equation governing the ball's position is obtained. The forces acting on the ball are:

- Gravitational force: $F_g = mg$

Where m is the mass of the ball.

- Magnetic force: $F_m = \frac{K_m i^2}{y^2}$

Where K_m is the electromagnet force constant and y is the vertical distance from the solenoid and the position of the ball.

Using Newton's second law:

$$m\ddot{y} = \frac{K_m i^2}{y^2} - mg$$

2.2.2 Solenoid Electrical Equation (RL)

Applying Kirchhoff's voltage law to the solenoid circuit:

$$u(t) = Ri + L \frac{di}{dt} \Rightarrow \dot{x}_3 = \frac{1}{L}(u(t) - Rx_3)$$

2.2.3 Complete Dynamic Model in State-Space Form and State Variable Definition

The system can be modeled in state-space form by defining the following state variables:

$$x_1 = y(t) \quad (\text{position with respect to the post}), \quad x_2 = i(t) \quad (\text{current}) \quad x_3 = \dot{y}(t) \quad (\text{velocity})$$

The state-space equations for the system are as follows:

$$\begin{cases} \dot{x}_1 = x_3 \\ \dot{x}_2 = \frac{1}{L}(u(t) - Rx_2) \\ \dot{x}_3 = \frac{1}{m} \left(\frac{K_m x_2^2}{(y_0 - x_1)^2} - mg \right) \end{cases} \quad (1)$$

Where y_0 is the maximum position at which the ball can remain. With measurable outputs:

$$\begin{cases} y_1 = x_1 \\ y_2 = x_3 \end{cases}$$

2.3 Estimation of the Parameters

Estimation of R and L

To estimate the resistance R and inductance L of the solenoid, the electrical dynamics were considered in isolation by removing it entirely, thereby eliminating the effects of the magnetic force. The solenoid's equivalent circuit is modeled as a series RL circuit, described by the following equation:

$$u(t) = Ri(t) + L \frac{di}{dt} \Rightarrow G(s) = \frac{1}{Ls + R}$$

By applying a step input signal $u(t)$, the current response $i(t)$ exhibits the behavior of a rising exponential function:

$$i(t) = i_{ss}(1 - e^{-t/\tau}), \quad \text{con} \quad \tau = \frac{L}{R}$$

Where:

- u_0 is the amplitude of the voltage step;
- τ is the time constant of the circuit;
- $i_{ss} = \frac{u_0}{R}$ is the steady-state value of the current.

Estimation of the resistance R :

- It was determined by calculating the ratio between the amplitude of the input step signal and the steady-state value of the resulting current:

$$R = \frac{u_0}{i_{ss}} = 9.76\Omega$$

Estimation of the inductance L :

- The inductance was calculated based on the time constant τ , which was estimated as the time required for the current to reach 63% of its final steady-state value:

$$i(\tau) = 0.63 \cdot i_{ss}$$

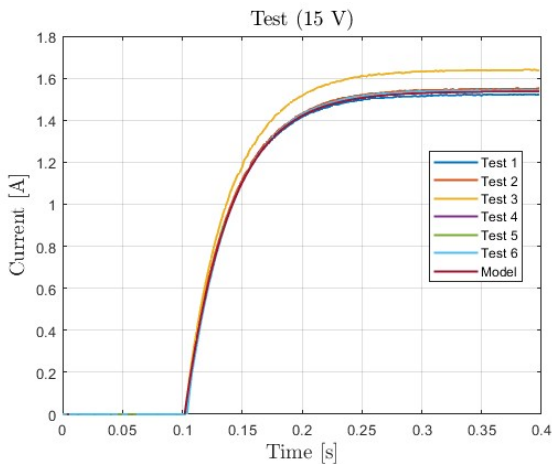
- Once the characteristic time constant was determined, it was multiplied by the resistance to obtain the inductance:

$$L = \tau \cdot R = 370.80\text{mH}$$

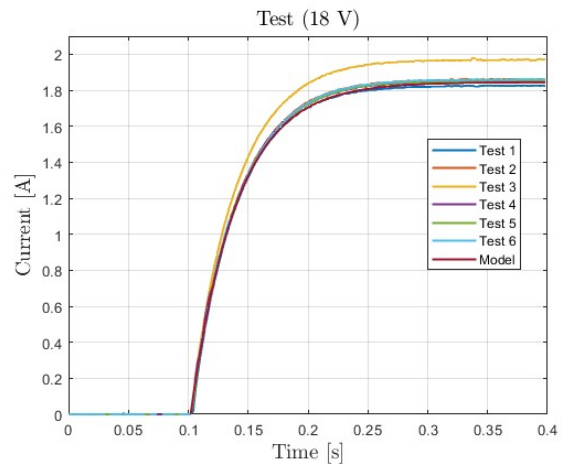
These 2 approaches were applied to multiple step response tests, and the resulting values were averaged to improve the robustness of the estimation. The results were subsequently validated in both the time and frequency domains.

Validation of the electric model

- **Time validation:** The simulated and real responses were evaluated through graphical plots in MATLAB.

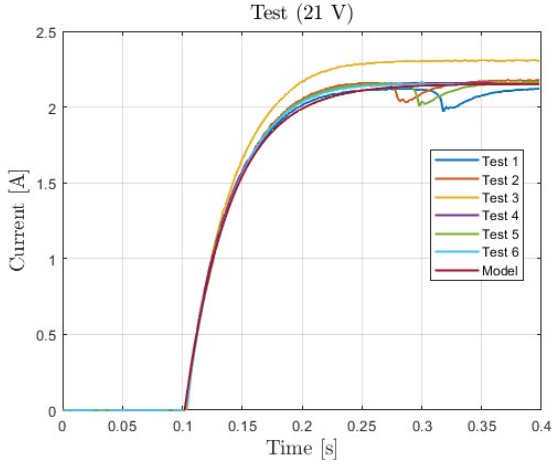


(a) Input step of 15V

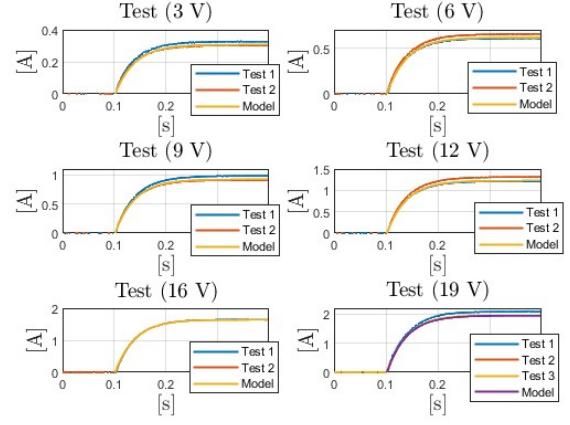


(b) Input step of 18V

Figure 4: Comparison between the experimental tests and the model



(a) Input step of 21V



(b) Different amplitudes of input steps

Figure 5: Comparison between the experimental tests and the model

It can be observed that the parameters of the model fit the experimental data well, with the data obtained both by applying steps of different amplitudes and by repeating steps of the same amplitude on different days, highlighting the redundancy in the reliability of our model.

- **Frequency validation:** A comparison was made between the Bode plot of the model and that of the experimental data obtained by inputting sinusoidal signals at different frequencies. The phase shift and amplitude were visually observed and then plotted on top of the model's Bode plot to ensure they fit. In the graph, the red markers represent the respective phase and magnitude values obtained experimentally.

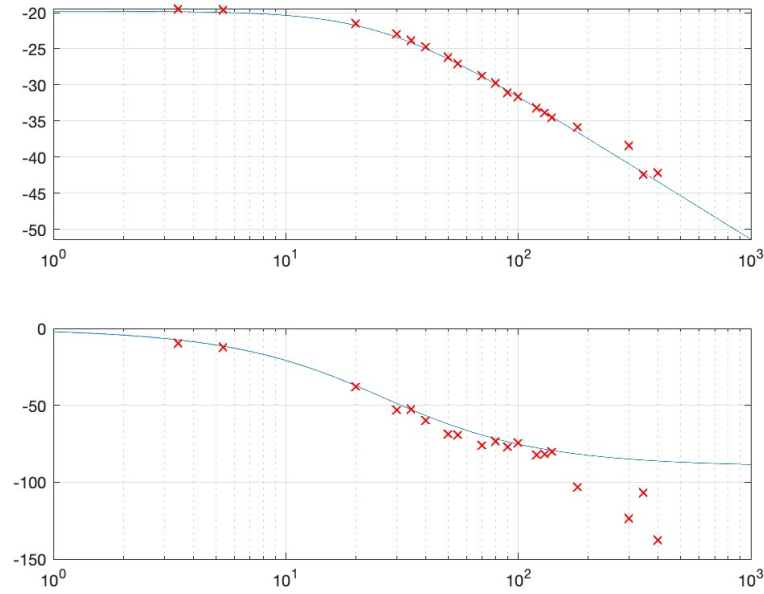


Figure 6: Frequency validation of the electric model

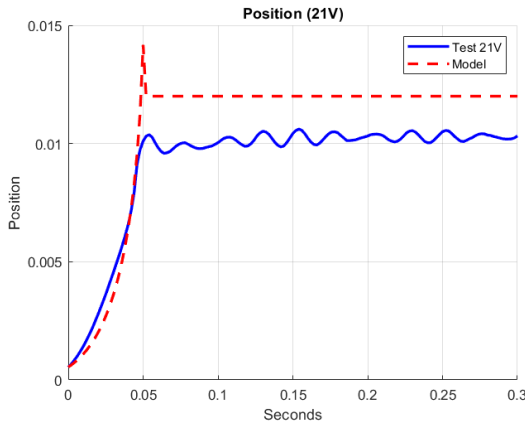
It is observed that our model fits the experimental data well at mid-low frequencies and around the pole frequency, but much less so at higher frequencies (above 100 rad/s). However, this issue is mitigated by selecting, during the design phase, the frequency of a future controller within an acceptable frequency range.

Estimation of the electromagnet force constant K_m

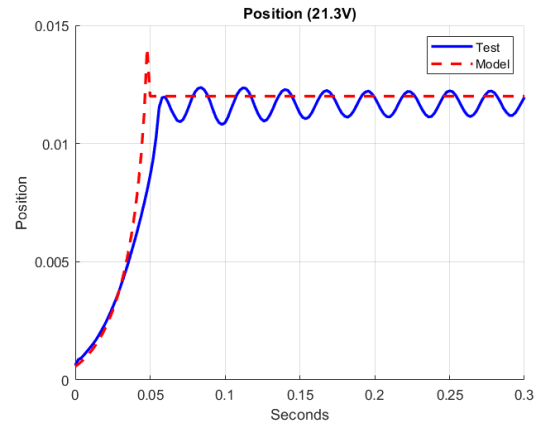
From the ball's dynamics, it can be deduced that:

$$m\ddot{y} = \frac{K_m i^2}{y^2} - mg \Rightarrow K_m = \frac{my^2(\ddot{y} + g)}{i^2} = 2,44 \times 10^{-5} \frac{\text{N} \cdot \text{m}^2}{\text{A}^2}$$

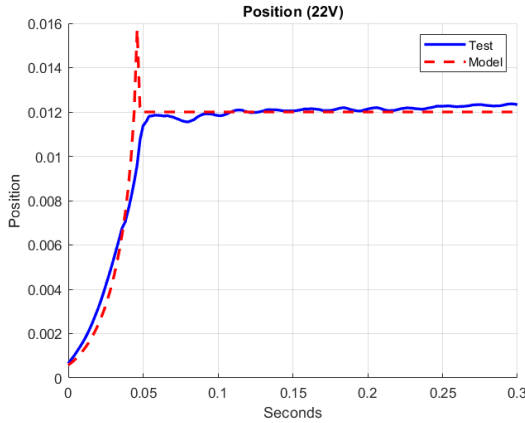
The sphere position model uses a fixed electromagnetic parameter. However, according to this model, the electromagnetic parameter changes depending on the air gap. To derive a suitable magnetic constant that best fits most of the tested values, we adopted the Curve Fitting algorithm. By evaluating the transient responses of different step tests, starting from the k_m value from the datasheet, we iteratively adjusted the magnetic constant to match the slopes of the considered transients. Various voltages were chosen where the ball is attracted by the electromagnet force, as this will be the region of interest for control. The tests were performed with step amplitudes of 21V, 21.3V, 21.6V, 22V, and 22.5V, from which the following response graphs were obtained:



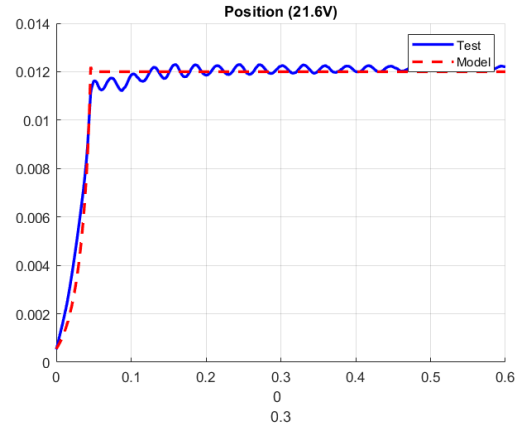
(a) K_m_{21V}



(b) $K_m_{21.3V}$



(c) K_m_{22V}



(d) $K_m_{21.6V}$

Figure 7: System responses to different step inputs

Conclusions

The developed nonlinear dynamic model:

- accurately represents the behavior of the system within the considered operational domain;
- includes both mechanical and electrical coupled components;
- has been experimentally validated, demonstrating good consistency between the model and the real system.

Electrical System Parameters	Measured/Estimated	Datasheet [1]
Resistance R [Ω]	9.76	10.00
Inductance L [mH]	370.80	412.50
Number of turns N_c	–	2450
Geometric Parameters		
Coil length l_c [mm]	–	82.50
Core radius r_c [mm]	–	8.00
Magnetic and Mechanical Parameters		
Electromagnet force constant K_m [$\text{N}^2\text{m}^2/\text{A}^2$]	2.44×10^{-5}	6.53×10^{-5}
Steel ball travel T_b [mm]	12	14
Steel ball mass M_b [Kg]	-	0.068
Steel ball radius r_b [m]	-	1.27×10^{-2}
Offsets and Sensors Sensitivities		
Current Sense Voltage Offset q_{CS} [V]	-0.0166×10^{-2}	-
Ball Position Sensor Voltage Offset q_{PS} [V]	-0.0231×10^{-2}	-
Ball Position Sensor Sensitivity K_b [m/V]	3.3×10^{-3}	2.8×10^{-3}

Table 1: Comparison between measured/stimated system parameters and datasheet values

3 Open Loop analysis:

3.1 Open Loop Analysis of the Electric System

In this section, the open-loop behavior of the electric model is analyzed, which consists of a first-order RL circuit is analyzed, represented by the following transfer function:

$$G(s) = \frac{1}{L_c s + R_{tot}} \quad (2)$$

where L_c is the inductance and R_{tot} is the total resistance of the system ($R_c + R_s$).

3.1.1 System Pole and Stability

The system has a single pole located at:

$$p = -\frac{R_{tot}}{L_c} = -26.317 \quad (3)$$

Since the pole is on the left half of the complex plane, the system is **asymptotically stable**.

3.1.2 Time-Domain Characteristics:

The time-domain response to a unit step input is evaluated using a MATLAB script. All performance indices related to the system's time response are assessed as follows:

- **Settling Time:** Time for the output to remain within 2% of the steady-state value.
 $t_s = 0.1486 \text{ s}$
- **Overshoot:** Maximum amount by which the output of a system exceeds its desired final (steady-state) value, expressed as a percentage of that final value. This system is a First order linear system, meaning it has no overshoot.

$$\text{Overshoot}_{\%} = 0$$

- **Time Constant:** The characteristic time constant is given by:

$$\tau = \frac{L_c}{R_{tot}} = 0.038 \text{ s}$$

- **Non Zero Steady State Error:** The Static Gain of the Open Loop system is $\gamma = \frac{1}{R_{tot}}$ resulting in a response of around 0.1 A to the unit step reference.

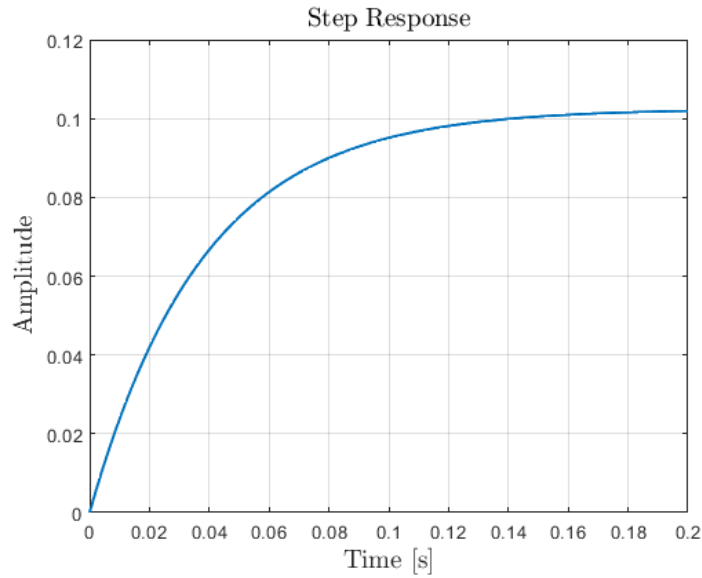


Figure 8: Response to a unit step input.

3.1.3 Frequency-Domain Analysis

It is here proposed the frequency domain analysis using the bode plot which enable to compute the following quantities:

- **Gain Margin (GM):** Increase in gain required to bring the system to instability. The Gain Margin is infinite because the open-loop Bode plot does not cross the -180° phase line; hence, there is no phase crossover frequency. As a result, no finite gain can make the system unstable in open loop.
- **Phase Margin (PM):** Amount of additional phase required to reach the critical value of -180° . Even here the phase margin is infinite, because the system gain never reaches 0 dB.
- **Crossover Frequencies:** Frequencies where the magnitude is 0 dB and the phase is -180° , respectively. As said before, there is no crossover frequency.

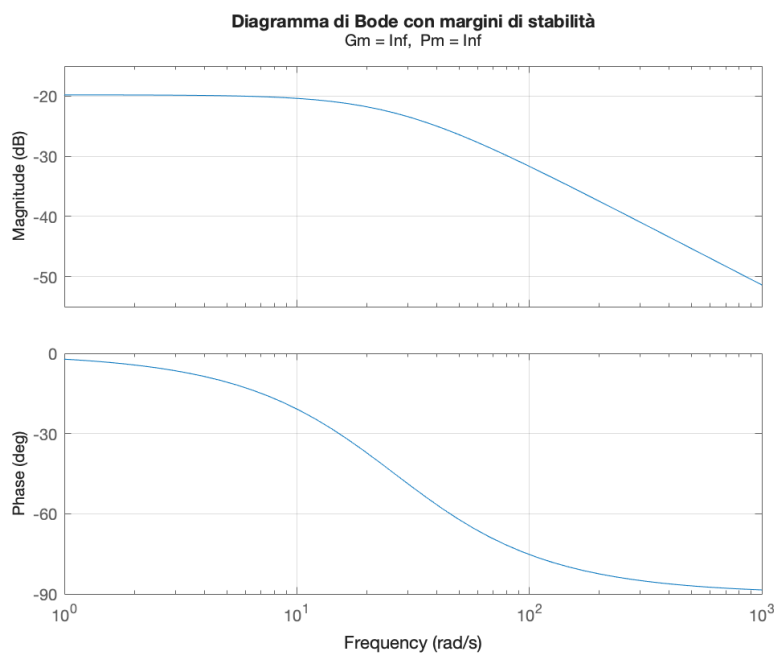


Figure 9: Bode Plot of the Electric System

3.1.4 Controllability and Observability

Give the system in its state space representation:

$$\dot{x}_2(t) = -\frac{R_{tot}}{L_c} \cdot x_2(t) + \frac{1}{L_c} \cdot u(t), \quad y_2(t) = x_2(t) \quad (4)$$

Using the controllability matrix:

$$\mathcal{C} = [B \quad AB \quad \dots \quad A^{n-1}B] = \left[\frac{1}{L_c} \right], \quad (5)$$

we find that the system is **fully controllable**, since $\text{rank}(\mathcal{C}) = n = 1$.

Using the observability matrix: $\mathcal{O}$

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \dots \\ C^{n-1}A \end{bmatrix} = \begin{bmatrix} 1 \\ -\frac{R_{tot}}{L_c} \end{bmatrix}$$

Since the matrix $\mathcal{O}$ is a 2×1 matrix, its rank is 1. Therefore, the system is **fully observable**, as the rank of the observability matrix equals $n = 1$, meaning the system is **fully observable**, based on the rank of the observability matrix.

3.1.5 Summary of Results

Metric	Value
Pole	-26.317
Time Constant (τ)	0.038
Overshoot	0%
Settling Time	0.1486 s
Gain Margin	$\infty \text{ dB}$
Phase Margin	$\infty \text{ degrees}$
Controllability	Full
Observability	Full

Table 2: Summary of open-loop analysis results for the electric model.

3.2 Open-loop Analysis of the Mechanical System:

The open-loop behavior of the mechanical system—comprising a steel ball balanced on a platform or track—is inherently unstable. In the absence of feedback control, any small displacement of the ball from its equilibrium position results in an unbounded divergence from that position over time. This is due to the fact that the ball experiences a restoring force in the wrong direction: rather than returning to equilibrium, gravity amplifies the displacement, causing the ball to accelerate further away. This instability necessitates the use of a closed-loop control strategy to actively stabilize the system.

4 Task 1: Electromagnet Coil Current Control

The first task involves designing a controller for the electrical model using frequency-domain techniques. The process begins with defining the controller specifications, followed by simulations on the ideal model. Finally, the controller is validated on the real system to assess its performance.

4.1 Coil Current Controller:

4.1.1 Selection of the Specifications

The controller is required to deliver enhanced performance to meet the requirements of the overall control system. This level of control allows the external loop to perceive the internal current control loop as having a near-unitary response, effectively simplifying the overall system dynamics. The design is guided by the following criteria:

1. **Zero Steady-State Error:** The system must accurately track the reference signal with zero steady-state error, ensuring precise current regulation.
2. **Overshoot:** The overshoot should be less than 6% to prevent excessive elongation or oscillations in the transient response and to maintain the output within acceptable bounds.
3. **Settling Time:** The settling time should be less than 0.1 seconds, allowing the system to reach steady-state rapidly and minimizing transient duration.
4. **Bandwidth:** Since in open loop the electrical linear model is no more reliable for frequencies over 100 rad/s, as it can be seen in Figure 6 a lower bandwidth is required.

4.1.2 Design of the Controller:

To fulfill the new performance specifications, the `rltool()` utility in MATLAB is employed. The design process involves shaping the root locus by imposing geometric constraints derived from the performance requirements, guiding the placement of dominant closed-loop poles.

The constraint on **Overshoot**, limited to less than 6%, once again corresponds to a minimum damping ratio $\zeta \approx 0.6$. This imposes an angular region in the complex s -plane defined by:

$$\theta = \cos^{-1}(0.6) \approx 51^\circ$$

This angle constrains the poles to lie within a conical region symmetric with respect to the negative real axis.

For the **Settling Time**, the requirement of $T_s < 0.1$ s leads to a higher natural frequency:

$$T_s \approx \frac{4}{\zeta \omega_n} < 0.1 \Rightarrow \omega_n > \frac{4}{0.6 \cdot 0.1} \approx 63.5 \text{ rad/s}$$

For these reasons, the requirements are rewritten as:

$$63.5 \frac{\text{rad}}{\text{s}} < \omega_n < 100 \frac{\text{rad}}{\text{s}} \tag{6}$$

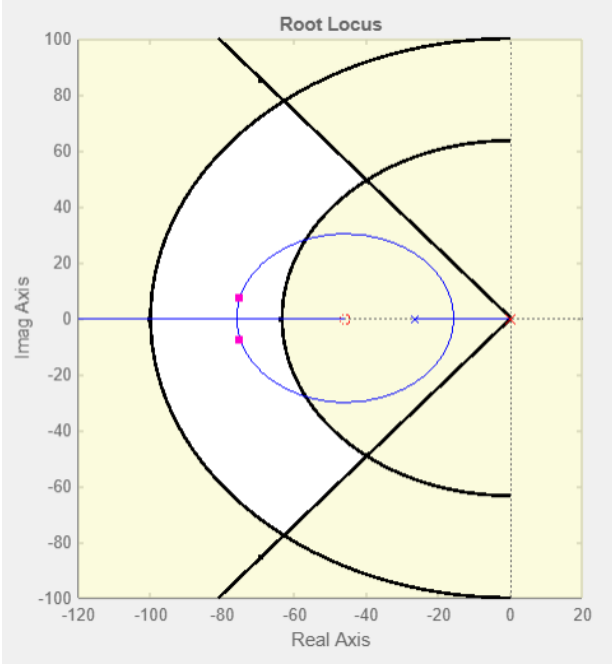
$$\zeta > 0.6 \tag{7}$$

To achieve **zero steady-state error**, the integral action is introduced.

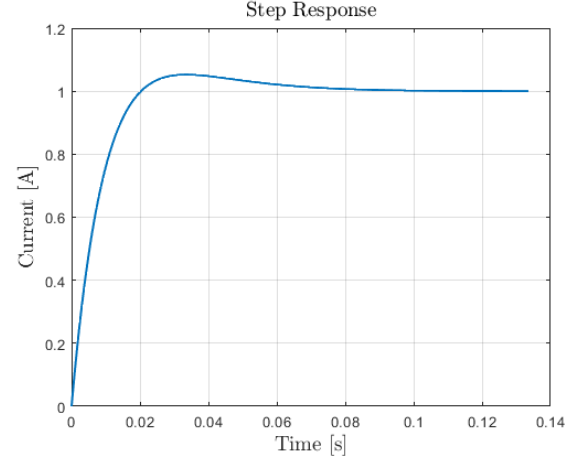
Once the region of interest is built, it is possible to build the controller. A **PI controller** is chosen for this specifications. The resulting controller is:

$$C(s) = \frac{29.924s + 1103.3}{s}$$

Figure 10 illustrates the target region of the poles for the controller as well as the actual closed-loop poles achieved.



(a) Root Locus of the Electric System



(b) Step Response of the Closed-Loop System

Figure 10: Root locus and step response of the system with the fast controller.

4.1.3 Antiwindup scheme

Input saturation due to voltage limitations requires the implementation of an anti-windup strategy to prevent the windup effect. Two main techniques were tested: Back-calculation and Clamping.

1. The **Back-Calculation technique** involves feeding back the difference between the saturated and unsaturated controller output to the integrator input. This feedback term helps to reduce the integrator state when the actuator is saturated, thus preventing integrator windup and improving system stability.
2. The **Clamping Method** disables the integrator action whenever the controller output reaches its saturation limits and the integrator would cause further increase in the output. By "clamping" the integrator, this technique prevents the integrator state from growing uncontrollably, avoiding the windup effect.

As shown in Figure 11, clamping exhibits the fastest settling time, while back-calculation is slower and more conservative. Therefore, to maintain a faster response, the clamping technique is applied.

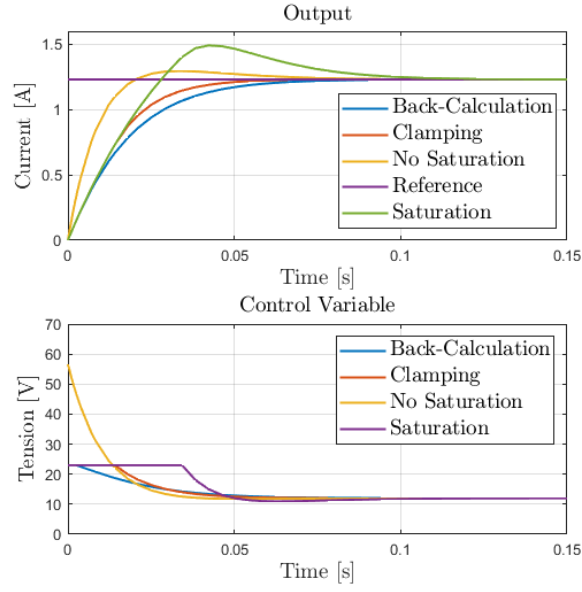
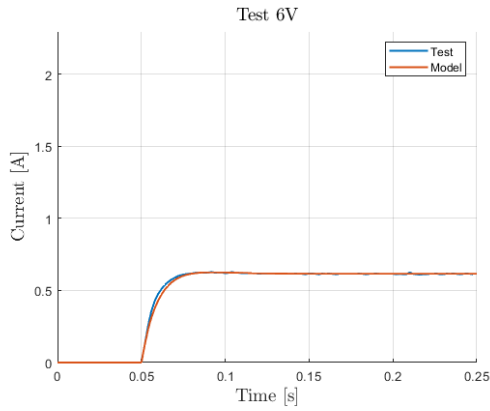


Figure 11: Antiwindup strategies comparison

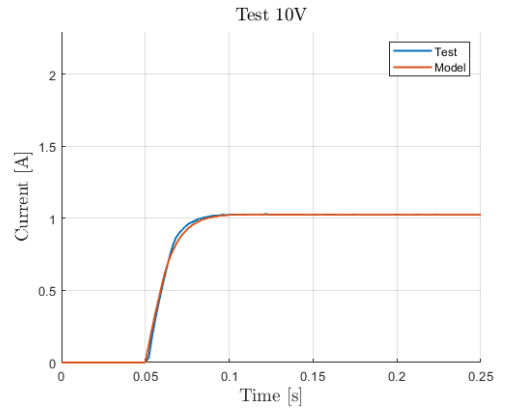
4.1.4 Validation of the Controller:

Time-Domain Validation:

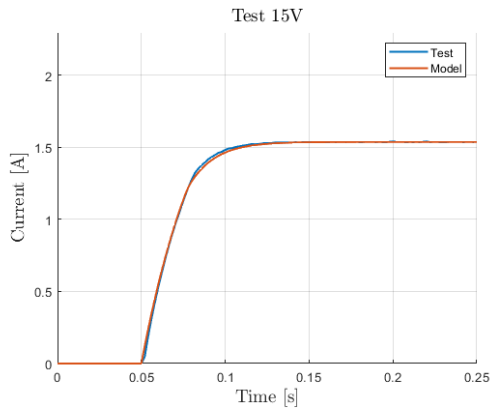
To validate the controller in the time domain, the coil current produced by the real system in response to different step input is measured. The plots in Figure 12 show a comparison between the simulated response of the model and the actual response of the real system also considering the antiwindup scheme.



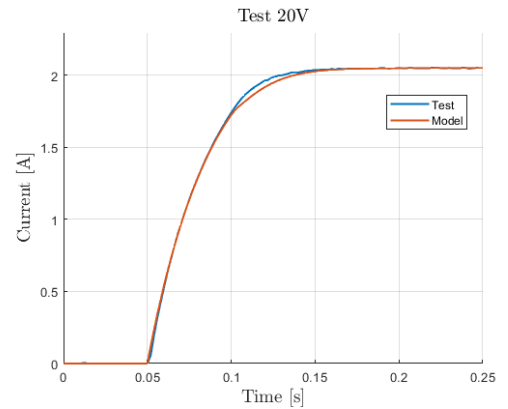
(a) Step Response with 6V



(b) Step Response with 10V



(c) Step Response with 15V



(d) Step Response with 20V

Figure 12: Step Responses

Frequency-Domain Validation:

To validate the controller in the frequency domain, the response of the real system is measured when subjected to sinusoidal inputs of varying frequencies. This allows assessment of the system's gain and phase behavior across the frequency spectrum.

The output coil current is recorded and compared to the reference sinusoidal signals, and the resulting magnitude and phase responses are compared with those predicted by the model; Figure 13 show this comparison, where the red crosses are the real data and the blue ones are from the simulation. However, as it can be noticed, the closed loop system is not able to follow the ideal bode diagram due to the non linearities introduced by the saturation and the windup strategy, but it is quite similar to the behaviour of the simulated model.

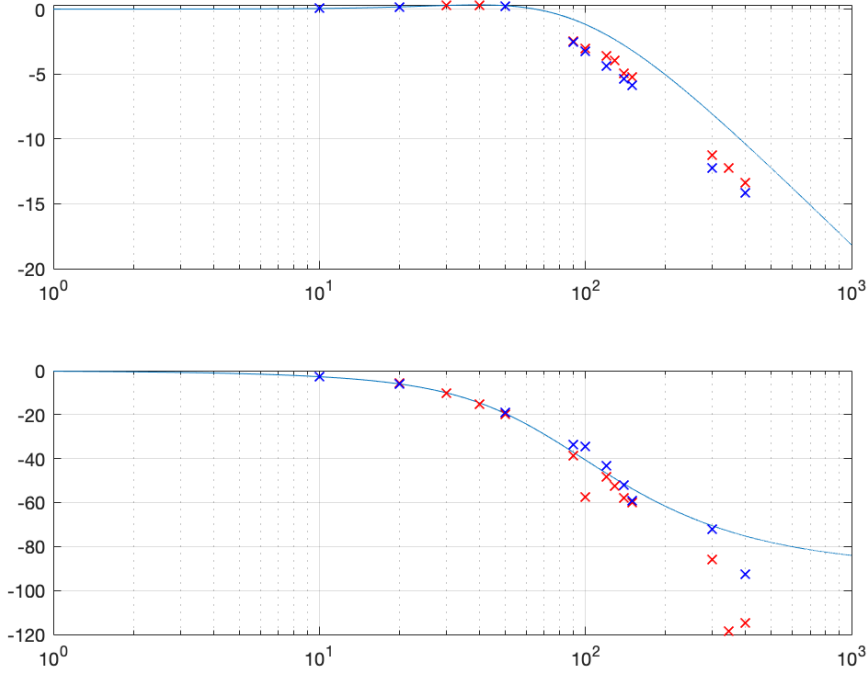


Figure 13: Bode Plot of the Closed-Loop System

4.1.5 Results:

Features	Value
Pole 1	$-72.2+7.44i$
Pole 2	$-72.2-7.44i$
Overshoot	5.25%
Settling Time	0.0612 s
Gain Margin	∞ dB
Phase Margin	153.44 degrees
crossover freq.	64 rad/s

Table 3: open-loop analysis results for the electric model with the designed controller.

4.2 Slow Coil Current Controller:

In order to overcome the problem relative to the frequency validation and the non linearities a slower controller is tested. Keeping the same overshoot limitation but with a smaller constraint about w_n with a requirement on the settling time lower than 0.5 seconds.

To satisfy the third requirement, namely a **settling time less than 0.5 seconds**, we use the following approximation:

$$T_s \approx \frac{4}{\zeta \omega_n} < 0.5 \text{ s}$$

Substituting the computed $\zeta = 0.63$, the required minimum natural frequency becomes:

$$\omega_n > \frac{4}{0.6 \cdot 0.5} \approx 12.7 \text{ rad/s}$$

the requirement are here summarized:

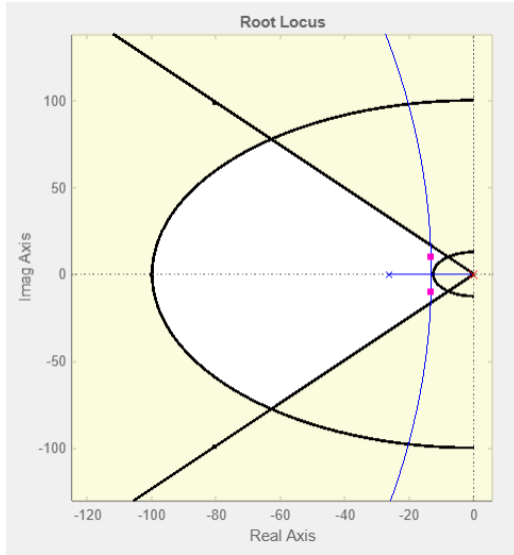
$$12.7 \text{ rad/s} < \omega_n < 100 \text{ rad/s} \quad (8)$$

$$\zeta > 0.6 \quad (9)$$

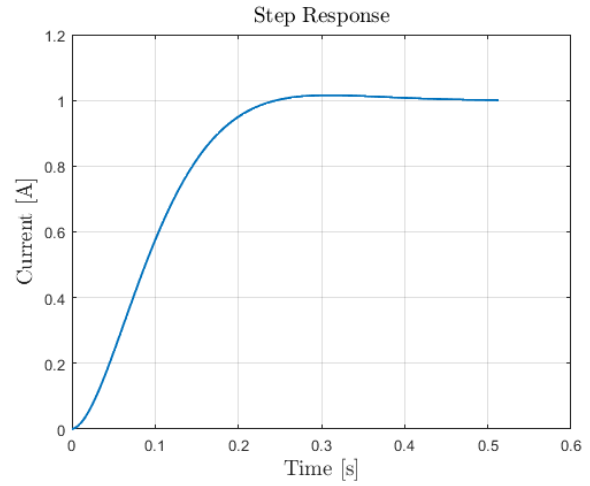
The final form of the slower controller is:

$$C(s) = \frac{0.15s + 104}{s}$$

Here are represented the root locus with the selected constraints and the step response:



(a) Root Locus of the Electric System



(b) Step Response of the Closed-Loop System

Figure 14: Root locus and step response of the system with the slow controller.

4.2.1 Validation of the Controller:

The last step involves the testing of the slower controller on the real system and the comparison with respect to the ideal model in the time domain and in the frequency domain.

Time Domain Validation:

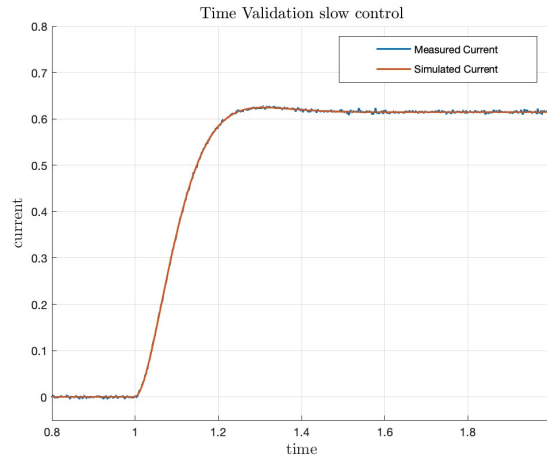


Figure 15: Time Validation slow control

Frequency Domain Validation:

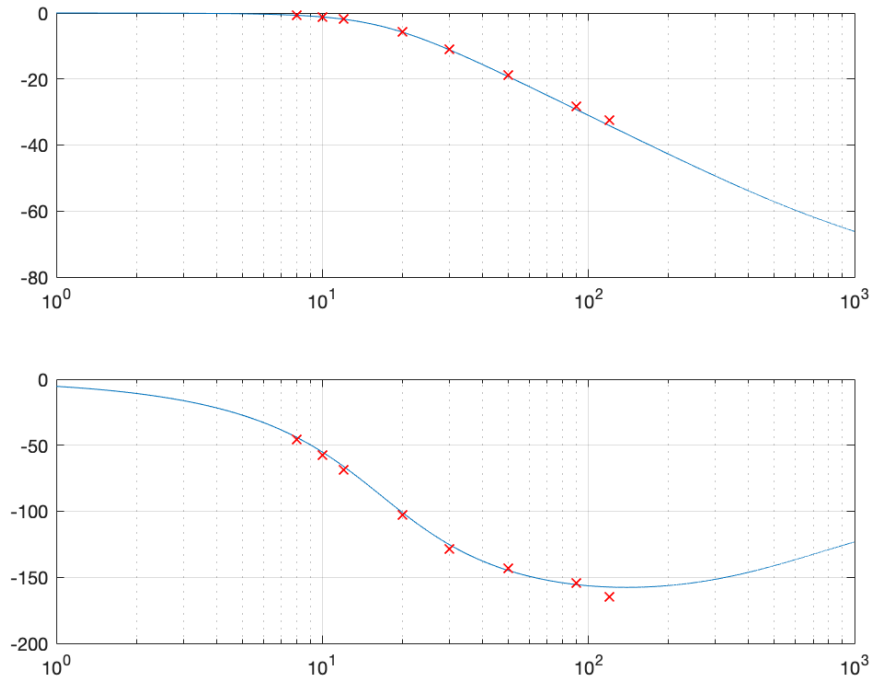


Figure 16: Bode Plot of the Closed-Loop System

4.2.2 Results:

Features	Value
Pole 1	$-13.3610 + 10.0975i$
Pole 2	$-13.3610 - 10.0975i$
Overshoot	1.57%
Settling Time	0.2218 s
Gain Margin	∞ dB
Phase Margin	180 degrees

Table 4: open-loop analysis results for the electric model with the first controller.

5 Task 2: Ball Position Stabilization (FB+SS)

5.1 Model Linearization Around the Equilibrium Point

In order to design a frequency-based and state space based controller, the nonlinear system describing the levitated sphere must be linearized around a defined operating point.

The steps to linearize the system are the following:

1. Consider the nonlinear model
2. Define the Equilibrium point
3. Linearize the system around the chosen point

Recall on the Nonlinear Model:

- **States:**

- x_1 : position of the sphere [m]
- x_2 : current through the coil [A]
- x_3 : vertical speed of the sphere [m/s]

- **Input:**

The only input to the system is the voltage applied to the electromagnet, denoted as u [V].

- **Output:** The measured quantities are :

- y_1 : position of the sphere [m]
- y_2 : current through the coil [A]

- **Nonlinear equation:** The nonlinear state-space model is in Equation 1

Definition of the Equilibrium Point:

The system is linearized around a nominal equilibrium position of the sphere, chosen arbitrarily as:

$$x_{1,eq} = 3.00 \text{ mm}$$

At this point, the steady-state values for the current, input voltage, and vertical speed are computed as:

$$x_{2,eq} = \sqrt{\frac{mg(y_0 - x_{1,eq})^2}{k_{mag}}} = 1.46 \text{ A}, \quad x_{3,eq} = 0 \text{ m/s}$$

$$u_{eq} = R_{tot} \cdot x_{2,eq} = 14.28 \text{ V}$$

Linearized State-Space Matrices

The nonlinear system is then linearized using Jacobian linearization, resulting in the following continuous-time state-space model:

$$\dot{x} = Ax + Bu, \quad y = Cx + Du$$

Where the state-space matrices A, B, C, D are:

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{R_{tot}}{L_c} & 0 \\ \frac{2k_{mag}x_{2,eq}^2}{m(y_0-x_{1,eq})^3} & \frac{2k_{mag}x_{2,eq}}{m(y_0-x_{1,eq})^2} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -26.31 & 0 \\ 2180 & 13.41 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{L_c} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2.69 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The output vector includes both position and current measurements, which are directly available from the sensors.

This linearized model is used in the frequency-based design of the outer-loop controller but also on the state space desing of the different controller using different techniques.

5.2 List of Specification for the Closed-Loop System

1. **Zero Steady-State Error:** The system must accurately track the reference signal with zero steady-state error, ensuring precise current regulation.
2. **Overshoot:** The overshoot should be less than 10-15% to prevent excessive elongation or oscillations in the transient response and to maintain the output within acceptable bounds.
3. **Settling Time:** The settling time should be less than 1 seconds.
4. **Damping ratio:** > 0.15
5. **Natural frequency:** 10 rad/s

5.3 Frequency-Based (FB) technique:

5.3.1 Design of the Frequency-Based Cascade Controller

The cascade control structure is designed to stabilize the levitated sphere around its equilibrium position. It consists of an inner current control loop and an outer position control loop. This architecture leverages the faster dynamics of the electrical subsystem to simplify the overall control design.

Inner Loop: Current Control

The inner loop is responsible for regulating the current in the electromagnet coil. It is designed first, as its dynamic response directly affects the performance of the outer position loop. The controller implemented here is the Coil Current Controller, previously validated and tested in Section 4.1.2:

$$C_{el}(s) = 29.924 + \frac{1130.3}{s}$$

This controller was selected after evaluating multiple alternatives for its superior performance. The closed-loop transfer function of the inner loop is then computed as:

- Electrical plant:

$$G_{el}(s) = \frac{1}{L_c s + R_{tot}} = \frac{1}{0.3708s + 9.7585}$$

- Current controller:

$$C_{el}(s) = K_p + \frac{K_i}{s} = 29.924 + \frac{1130.3}{s} = \frac{29.924s + 1130.3}{s}$$

Therefore, the closed-loop transfer function is:

$$G_{\text{el_close}}(s) = \frac{29.924s + 1130.3}{0.3708s^2 + 39.6825s + 1130.3}$$

This confirms the expected result and validates the implementation of the inner loop controller. **Outer**

Loop: Position Control

Once the inner loop is closed, the resulting system seen by the outer loop becomes the cascade of the closed-loop current dynamics and the magnetic actuation plant $G(s)$. The combined plant is:

$$G_2(s) = G_{\text{el_close}}(s) \cdot G(s)$$

The magnetic plant $G(s)$ is obtained by linearizing the nonlinear magnetic dynamics around the operating point. This linear model enables the design of the position controller.

The selected outer-loop controller is a phase-lead compensator given by:

$$C(s) = \frac{28.913(s + 34.87)}{s}$$

The following figures show the root locus with design constraints and the step response of the overall cascade system:

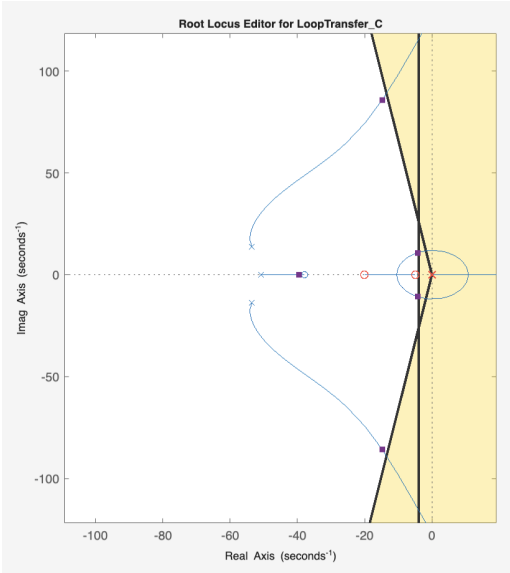


Figure 17: Root locus of the outer loop with design constraints.

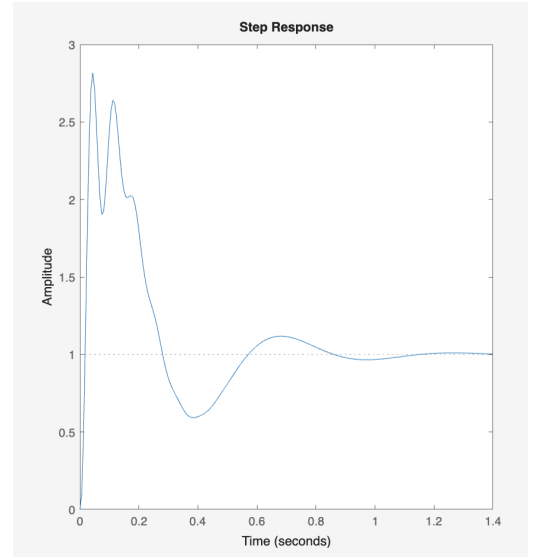


Figure 18: Step response of the full cascade control system.

It is worth noting that the overshoot constraint could not be strictly met without compromising other, higher-priority requirements. Nevertheless, the selected controller significantly improves the transient response and ensures zero steady-state error in the position. The combination of fast inner-loop current control and anticipatory position compensation results in a robust and well-performing closed-loop system.

5.3.2 Time Validation:

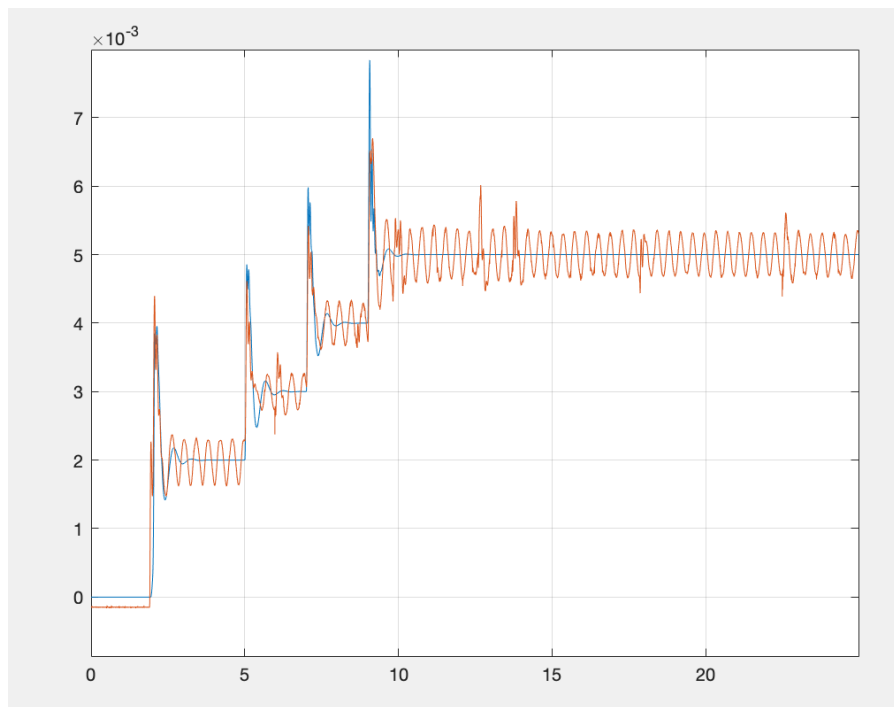


Figure 19: Time Validation of Cascade Controller

5.3.3 Frequency Validation:

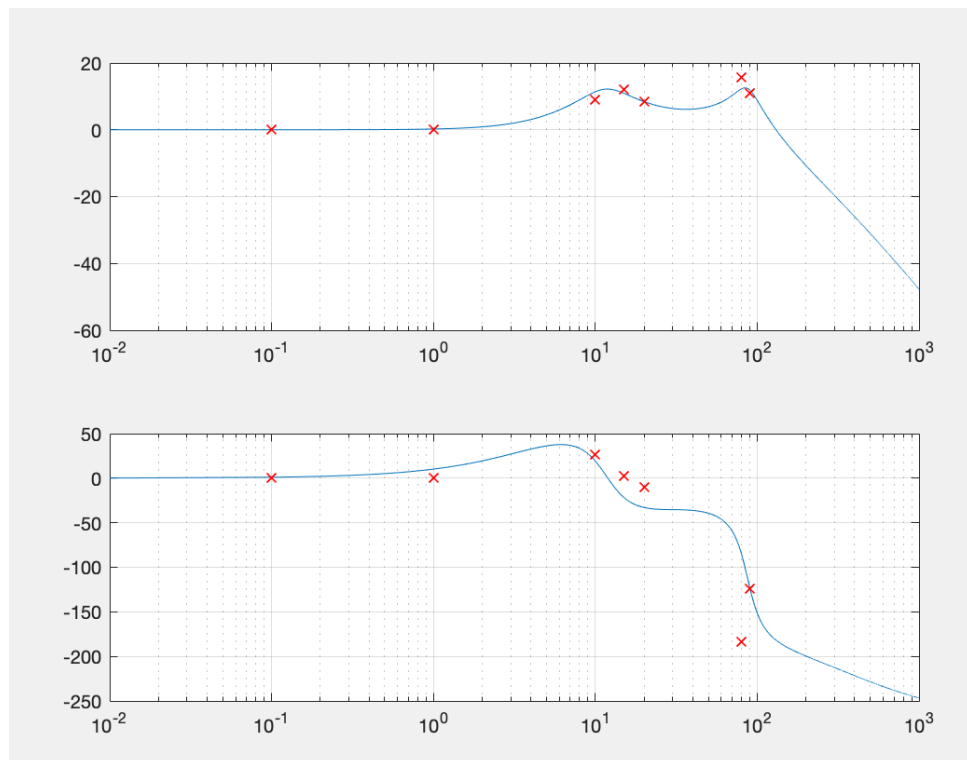


Figure 20: frequency Validation of the Cascade

5.4 State-Space (SS) techniques

State-Space control strategies are analyzed with the aim of stabilizing the ball at its equilibrium position and enabling it to track a position reference.

5.4.1 Pole Placement

The design of a controller for this system requires advanced methods capable of ensuring stability, good dynamic performance, and zero steady-state error. In this context, a design approach based on local linearization of the system and state feedback control via pole placement was adopted. A nonlinear magnetic levitation system is considered, described by nonlinear differential equations governing the dynamics of position and current.

To make the system tractable using linear control methods, the linearized system, previously computed is used.

The linearized form of the system is:

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \quad (10)$$

where:

- $x(t) \in \mathbb{R}^n$ is the state vector,
- $u(t) \in \mathbb{R}^m$ is the input vector,
- $y(t) \in \mathbb{R}^p$ is the output vector,
- $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, and $C \in \mathbb{R}^{p \times n}$ are the matrices of the linearized system.

Verification of Controllability and Observability

For pole placement to be applicable, the system must be both fully controllable and observable. To verify these properties, the following matrices are constructed based on the system matrices:

The **Controllability Matrix**:

$$\mathcal{C} = [B \quad AB \quad A^2B] = \begin{bmatrix} 0 & 0 & 36.0729 \\ 2.69 & -70.7639 & 1861.1088 \\ 0 & 36.0729 & -949.8220 \end{bmatrix}$$

Then:

$$\text{rank}(\mathcal{C}) = 3$$

The **Observability Matrix**:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -26.31 & 0 \\ 2180 & 13.41 & 0 \\ 0 & 692.5561 & -26.31 \end{bmatrix}$$

Then:

$$\text{rank}(\mathcal{O}) = 3$$

Since both $\mathcal{C}$ and $\mathcal{O}$ have full rank (3), the system is **controllable** and **observable**. Only in this case is it guaranteed that the state can be controlled through a suitable input $u(t)$ and reconstructed from partial measurements $y(t)$.

State Feedback Controller Design (Pole Placement) with Error Integrator

Once controllability has been verified, a state feedback controller of the form:

$$u(t) = -Kx(t)$$

is designed. This feedback modifies the system dynamics as follows:

$$\dot{x}(t) = (A - BK)x(t)$$

The goal is to place the eigenvalues (poles) of the matrix $(A - BK)$ in locations in the complex plane that ensure stability and the desired dynamic performance (e.g., settling time, overshoot, rise time). The poles are selected to lie entirely in the left-half complex plane (i.e., with negative real parts), preferably real or with limited imaginary parts to avoid unwanted oscillations. In our case, since the system is controllable, a state feedback controller is designed as follows:

$$u(t) = -K_{pp}x(t)$$

The System must be Extended, by introducing an integral state, to improving control performance and eliminating steady-state error. The system should satisfy the following conditions:

1. Constraint on the number of inputs and outputs

In the considered system, the number of inputs is $m=1$ while the number of outputs is $p=2$. This configuration, where $p > m$, does not satisfy the general condition

$$p \leq m$$

which is necessary for the integral state extension to be applied directly on the entire output vector. Indeed, having more outputs than inputs implies that it is not possible to guarantee simultaneous integral regulation of all outputs without encountering overdimensioning issues or infeasibility in designing the integral controller.

Therefore, to apply the system extension, it is necessary to limit the integral action to a subset of outputs — the first output corresponding to $C(1,:)$ — or to design a more complex control strategy to handle the mismatch between inputs and outputs.

2. Absence of derivative actions

The linearized system does not exhibit control actions based on the derivatives of inputs or outputs, as evidenced by the structure of matrices A and B , which contain no explicit derivative terms with respect to the control variables. This ensures that the introduction of the integral state is compatible with the system dynamics and that the integrator can effectively act on steady-state error without causing instability or undesirable effects due to derivative components.

Given the characteristics of the system, the integral state extension is applicable provided that the outputs subject to integration are properly selected, thereby ensuring controllability of the extended system and effectiveness of the integral regulation. The absence of derivative actions facilitates the stability and synthesis of the integral controller.

To ensure zero steady-state error in the presence of constant inputs or disturbances, an integrator on the error is introduced:

$$\eta(t) = \int (r(t) - y(t)) dt$$

The addition of the integrator leads to the definition of an extended system with the augmented state:

$$\tilde{x}(t) = \begin{bmatrix} x(t) \\ \eta(t) \end{bmatrix} \in \mathbb{R}^4$$

and extended matrices:

$$\tilde{A} = \begin{bmatrix} A & 0 \\ -C_1 & 0 \end{bmatrix}, \quad \tilde{B} = \begin{bmatrix} B \\ 0 \end{bmatrix}, \quad M = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

where $C_1 = C(1, :)$ selects the first output (position).

Extended Controller Design with Integral Action

The desired poles for the extended system are selected as:

$$\lambda_{cl} = \{-5, -40, -50, -70\} \quad (11)$$

Choosing a pole closer to the origin for the integrator (e.g., -5) ensures that the integral action is slower than the main dynamics, avoiding instability. The choice of the poles was made to satisfy the dominant pole approximation, by placing the closest pole to the origin at -5 , while the remaining poles were placed one decade farther to the left in the complex plane. This ensures compliance with the settling time specification, as the settling time can be approximated by $t_a = \frac{4}{|\text{Re}(\text{dominant pole})|}$. Furthermore, the poles were selected to be real and negative, thereby avoiding complex conjugate poles that could lead to unwanted overshoot due to oscillatory transients. The pole placement technique is applied to obtain the extended gain matrix:

$$K_{en} = \text{place}(\tilde{A}, \tilde{B}, \lambda_{cl}) \implies K_{en} = [K_x \quad K_\eta] = [14959.56 \quad 51.42 \quad 311.79 \quad -19349.02]$$

with

$$K_x \in \mathbb{R}^{1 \times 3} \quad (\text{gains on the original states}), \quad K_\eta \in \mathbb{R} \quad (\text{gain on the error integrator}).$$

The control law becomes:

$$u(t) = -K_x x(t) - K_\eta \eta(t)$$

Conclusions

The resulting system includes:

- a state feedback controller designed to place the poles at $\{-40, -50, -70\}$;
- an integrator to eliminate steady-state error, with a pole at -5 ;

This architecture ensures:

- internal stability of the controlled system;
- fast and well-damped dynamic response;
- elimination of steady-state error;

The adopted procedure allows a comprehensive approach to designing a robust controller for unstable and nonlinear systems. The use of classical state-space control techniques, integrated with modern elements like integral action, guarantees high performance and robustness in the presence of disturbances and model uncertainties.

Experimental Validation

The state-space controller was designed using the pole placement technique applied to the linearized system model extended with an integral state. To verify the effectiveness and robustness of the designed controller, an experimental validation was conducted in the laboratory, consisting of both time-domain and frequency-domain analyses.

Time-Domain Validation

A step input was applied to the system to observe its transient response. The measured output was compared against the expected reference signal to evaluate the controller's ability to track the desired setpoint and to reject disturbances. Key performance indicators such as settling time, overshoot, and steady-state error were analyzed, confirming the proper operation of the controller and the successful elimination of steady-state errors due to the integral action.

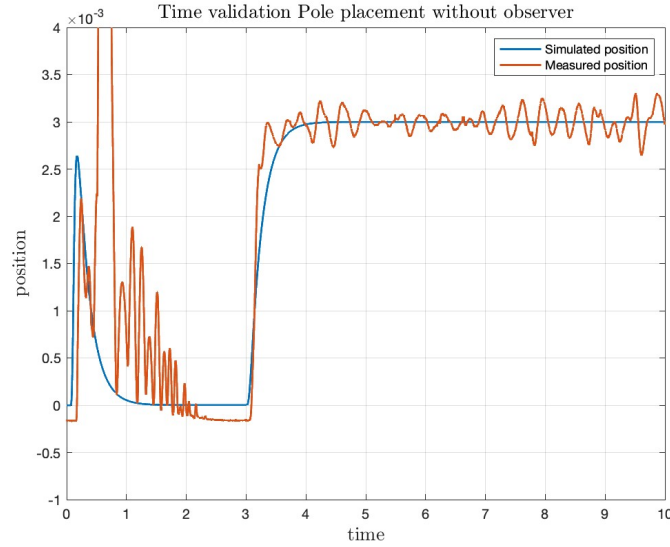


Figure 21: PP Time Validation

Frequency-Domain Validation

To further assess the controller's frequency response and robustness, sinusoidal inputs of varying frequencies were applied to the system. The resulting output signals were recorded, and the gain and phase shift relative to the input were observed superimposing the the measured output with the signal. These data were used to plot the Bode diagram of the closed-loop transfer function, displaying both the amplitude and phase characteristics over the frequency range of interest. The Bode plots provided insight into the stability margins and bandwidth of the controlled system, confirming that the experimental frequency response aligns closely with the theoretical predictions derived from the state-space model and the designed controller. This comprehensive validation ensures that the controller performs as intended under practical operating conditions.

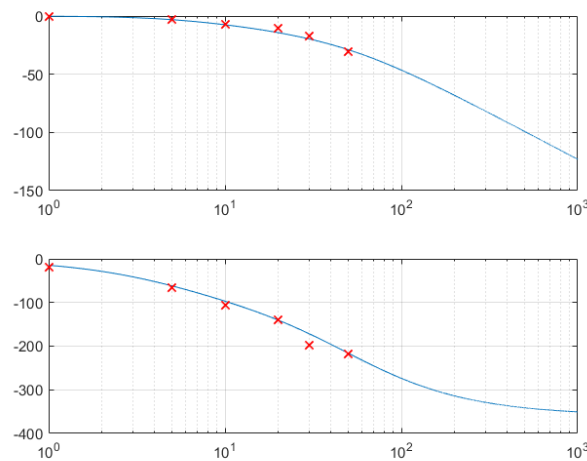


Figure 22: Frequency Validation of PP

Overall, the combination of time-domain step response and frequency-domain sinusoidal testing constitutes a rigorous approach to validate the pole placement based state-space controller in both transient and steady-state regimes.

5.4.2 Linear Quadratic Control

Another method used to stabilize the system and ensure that the position tracks a reference signal is the Linear Quadratic Control (LQ). LQ is based on the previously linearized system; however, since the task also requires the position of the ball to follow a reference, the system must be augmented by introducing two additional states. These states represent the integrals of the tracking errors between the references and the corresponding outputs (position and current).

To augment the system, the following conditions must be verified:

- The number of inputs must be greater than or equal to the number of outputs.
- There must be no derivative actions in the system.

The second condition is satisfied because the transfer function of the linearized system contains no zeros. However, the first condition is not fulfilled, as the system has two outputs and one input. Consequently, it is only possible to augment the system with one integral state corresponding to the tracking error of a single output. In this specific case, we choose the output for which we want the tracking error to asymptotically reach zero. Before augmenting the system, it is necessary to verify that the “reduced” system has no zeros at the origin. The reduced system in state-space form is described using the previously computed matrices of the linearized system, allowing to compute the following transfer function:

$$G(s) = \frac{36.18}{s^3 + 26.32s^2 - 2180s - 5.737 \times 10^4}$$

As shown, $G(s)$ has no zeros at the origin.

The augmented system matrices are:

$$A_{enlarged} = \begin{bmatrix} A & 0 \\ -C(1,:) & 0 \end{bmatrix}, \quad B_{enlarged} = \begin{bmatrix} B \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

In LQ control, selecting the matrices Q and R is crucial. Q penalizes deviation of the states from the desired values, while R penalizes excessive control effort. Since the state variables have different physical units, we proceed using normalized matrices Q_n and R_n . Q is a diagonal matrix with elements chosen to assign:

- High priority to position, which is critical for control;
- High priority to velocity, since it should be zero at equilibrium;
- Low priority to current and the augmented state.

R is chosen to prevent the control input from becoming too aggressive, considering the saturation limits of the amplifier.

After tuning, the matrices are:

$$Q = \begin{bmatrix} 0.1 & 0 & 0 & 0 \\ 0 & 0.001 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.001 \end{bmatrix}, \quad R = 0.1$$

For better understanding the matrices, we multiply for a scaling factor in order to obtain:

$$Q = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 500 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R = 10$$

Since we are using a normalized approach, we need the maximum values of each state and the input, taken from the datasheet or experimental data:

$$x_{max} = [0.012 \quad 3 \quad 0.1 \quad 1], \quad u_{max} = 23$$

The original Q and R matrices are computed as:

$$Q_n = \begin{bmatrix} 6.944 \times 10^5 & 0 & 0 & 0 \\ 0 & 0.11 & 0 & 0 \\ 0 & 0 & 5 \times 10^4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R_n = 0.0189$$

Finally, to apply the LQ control, we must verify:

- The pair $(A_{enlarged}, B_{enlarged})$ is controllable;
- The pair $(A_{enlarged}, C_q)$ is observable.

The controllability matrix is:

$$M_R = \begin{bmatrix} 0 & 0 & 36.1775 & -952.0995 \\ 2.6969 & -70.9747 & 1868 & -4.9158 \times 10^4 \\ 0 & 36.1778 & -952.0995 & 1.0392 \times 10^5 \\ 0 & 0 & 0 & -36.175 \end{bmatrix}$$

Its rank is 4, indicating the system is controllable.

The matrix C_q is a scaled version of Q_n :

$$C_q = \begin{bmatrix} 833.33 & 0 & 0 & 0 \\ 0 & 0.333 & 0 & 0 \\ 0 & 0 & 223.6068 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The observability matrix O is:

$$O = \begin{bmatrix} 888.333 & 0 & 0 & 0 \\ 0 & 0.333 & 0 & 0 \\ 0 & 0 & 223.6068 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 833.33 & 0 \\ 0 & -8.7725 & 0 & 0 \\ 2.8746 \times 10^5 & 2.9996 \times 10^3 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 1.8167 \times 10^6 & 1.1179 \times 10^4 & 0 & 0 \\ 0 & 230.8689 & 0 & 0 \\ 0 & -7.8942 \times 10^4 & 4.8746 \times 10^5 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -2.942 \times 10^5 & 1.8111 \times 10^6 & 0 \\ 0 & -6.0759 \times 10^3 & 0 & 0 \\ 1.0627 \times 10^9 & 8.6167 \times 10^6 & 0 & 0 \\ -2180 & -13.4146 & 0 & 0 \end{bmatrix}$$

The rank of O is 4, confirming the system is observable.

With the following choice of matrices, using the command

$$K_{LQ} = \text{lqr}(A_{enlarged}, B_{enlarged}, Q_n, R_n) = [3.33885 \times 10^4, 90.6155, 970.2311, -8.2065 \times 10^4],$$

the first three components of the gain vector correspond to the system states, while the fourth component refers to the augmented state. Using this gain matrix, the resulting closed-loop poles are:

$$\text{closed_loop_poles} = [-0.0036, -8.7246, -1.2081 \times 10^2 + 1.0905 \times 10^2 i, -1.2081 \times 10^2 - 1.0905 \times 10^2 i].$$

However, these poles are not sufficient to meet the desired specifications and performance require-

ments of the task. In order to ensure that the closed-loop poles lie to the left of a specific bound on the complex plane (i.e., have sufficiently negative real parts to guarantee a desired rate of convergence), the system matrix was shifted by adding $3I_4$ to ensure the poles to have a real part lower than -3, where I_4 is the 4×4 identity matrix. The new gain matrix was computed as:

$$K_{LQ} = \text{lqr}(A_{\text{enlarged}} + 3I_4, B_{\text{enlarged}}, Q, R) = [3.0156 \times 10^4, 88.6672, 938.4869, -5.4898 \times 10^4],$$

yielding the following closed-loop poles:

$$\text{closed_loop_poles} = [-6, -12.1663, -1.2364 \times 10^2 + 1.0918 \times 10^2 i, -1.2364 \times 10^2 - 1.0918 \times 10^2 i].$$

Time-Domain Validation

To evaluate the steady-state error, overshoot, settling time, and to assess whether the experimental results are consistent with the simulated ones, a step reference input was applied at $t=3\text{s}$, with a final value equal to the equilibrium position. The resulting responses from both the simulation and the experimental system were then compared.

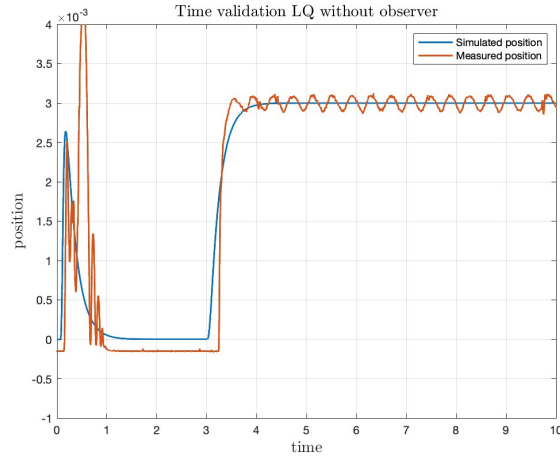


Figure 23: LQ Time Validation

Frequency-Domain Validation

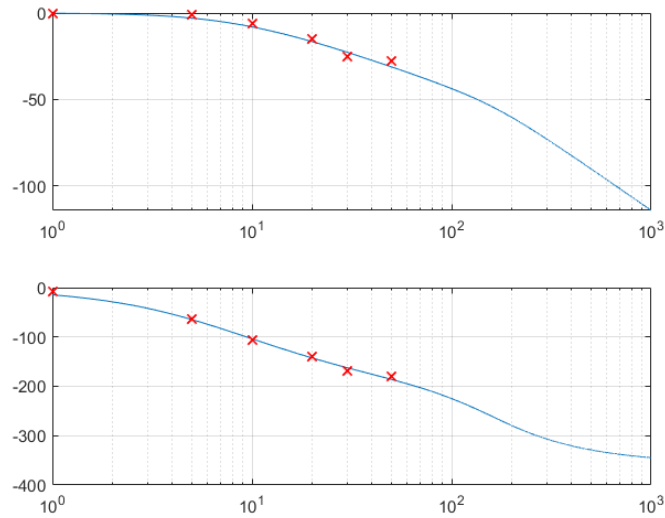


Figure 24: Frequency validation of LQ

5.4.3 Observer

Luenberger Observer Design

To ensure full implementability, a Luenberger observer was also designed to estimate the unmeasurable states, along with an error integrator to eliminate steady-state error. Since not all states are directly measurable, a Luenberger observer is used. It allows the estimation of the state $x(t)$ based on the measurement of the output $y(t)$ and the known input $u(t)$. The observer dynamics are:

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L(y(t) - C\hat{x}(t))$$

where $\hat{x}(t)$ is the state estimate and $L \in \mathbb{R}^{n \times p}$ is the observer gain matrix. The estimation error dynamics, defined as

$$e(t) = x(t) - \hat{x}(t),$$

are governed by:

$$\dot{e}(t) = (A - LC)e(t).$$

Therefore, the observer poles (eigenvalues of $A - LC$) can be placed in desired locations, faster than those of the controlled system, to ensure rapid convergence of the estimation error. The observer poles are chosen to be faster than those of the controller to ensure quick convergence:

$$\lambda_{obs} = \{-400, -500, -700\}$$

The observer gain matrix is computed as:

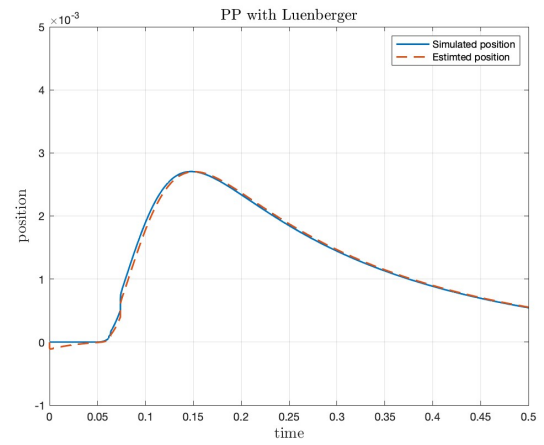
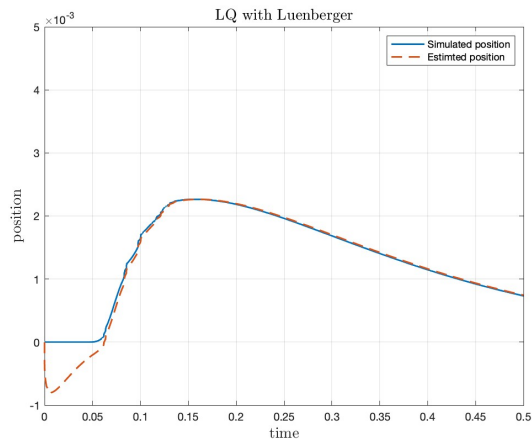
$$L = \text{place}(A^T, C^T, \lambda_{obs})^T$$

The observer dynamics are given by:

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L(y(t) - C\hat{x}(t))$$

Simulation-Based Evaluation of the Luenberger Observer with Different Controllers

The designed Luenberger observer was evaluated through numerical simulations in conjunction with two distinct state-feedback controllers applied to the linearized magnetic levitation system: one designed via the Pole Placement (PP) technique, and the other via the Linear Quadratic Regulator (LQR) approach. The observer's role is to provide real-time estimates of the system states, which are not all directly measurable, enabling full state-feedback implementation. Both controllers were integrated with the observer to assess the overall closed-loop performance in terms of transient response, robustness, and convergence speed of the state estimates. Simulation results demonstrate that the Luenberger observer paired with the Pole Placement controller exhibits superior performance compared to the observer combined with the LQR controller. Specifically, the PP-based controller allows for explicit placement of closed-loop poles, facilitating the design of a faster and more aggressive dynamic response tailored to the system requirements. This leads to quicker convergence of the observer estimation error and a more precise and rapid stabilization of the system states. In contrast, the LQR controller optimizes a quadratic cost function balancing control effort and state regulation, often resulting in a more conservative controller with slower transient dynamics. Consequently, the observer estimation error convergence and overall system response under LQR control are comparatively slower, although potentially more robust to model uncertainties. The ability of Pole Placement to directly assign pole locations, combined with the fast observer poles, ensures rapid error decay and enhanced performance in state estimation and control, which is critical for the magnetic levitation system's stability and responsiveness. These findings highlight the practical advantages of the Pole Placement approach when combined with a Luenberger observer in applications requiring fast and precise state estimation and control.



Time-Domain Validation for Luenberger + Pole Placement

A step input was applied to the system to check the performance based on real input signal.

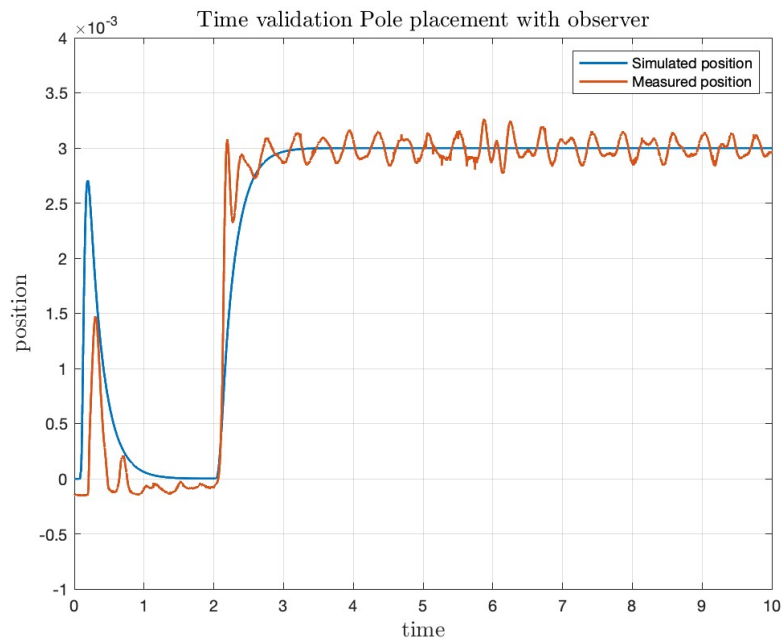


Figure 26: Pole Placement with Luenberger observer Time Validation

Frequency-Domain Validation

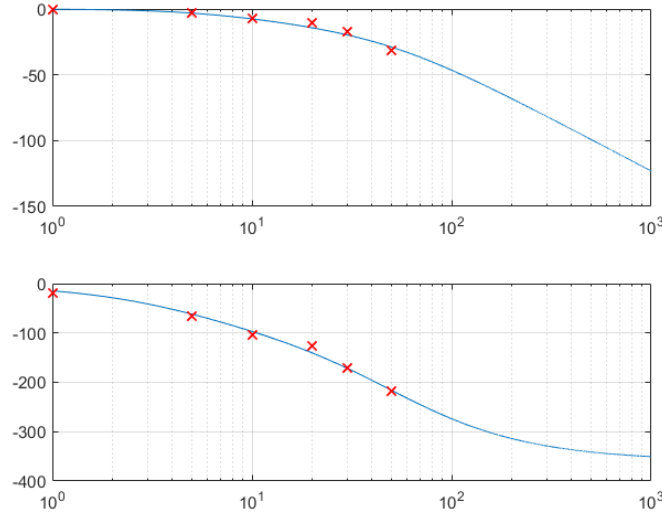


Figure 27: Frequency Validation of PP with Luenberger observer

Kalman Filter Design

In addition to the Luenberger observer, a Kalman Filter (KF) was designed for the estimation of the system states in the linearized magnetic levitation setup. The Kalman Filter is an optimal observer that provides the best linear unbiased estimate of the system states in the presence of process and measurement noise, assuming these noises are Gaussian with known covariance. Consider a continuous-time linear system affected by process and measurement noise:

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t), \quad y(t) = Cx(t) + v(t)$$

where:

- $x(t)$ is the state vector,
- $u(t)$ is the control input,
- $y(t)$ is the measured output,
- $w(t) \sim \mathcal{N}(0, Q)$ is the process noise,
- $v(t) \sim \mathcal{N}(0, R)$ is the measurement noise.

The Kalman Filter estimates the system state $\hat{x}(t)$ by minimizing the mean square estimation error. The dynamics of the observer are given by:

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L_{\text{KF}}(y(t) - C\hat{x}(t))$$

where L_{KF} is the Kalman gain matrix computed to minimize the steady-state error covariance. The estimation error $e(t) = x(t) - \hat{x}(t)$ evolves according to:

$$\dot{e}(t) = (A - L_{\text{KF}}C)e(t) + w(t) - L_{\text{KF}}v(t)$$

To ensure convergence and optimality, the Kalman Filter requires the system to be:

- **Observable:** The pair (A, C) must be observable so that the states can be inferred from the outputs.
- **Reachable:** The pair (A, B_q) must be **controllable**, where $Q = B_q B_q^T$ is a factorization of the process noise covariance matrix.

Selection of Covariance Matrices Q and R

In this implementation, the process noise covariance matrix $Q \in \mathbb{R}^{n \times n}$ was experimentally derived from data collected during operation at a constant input of 21V. The residuals between the measured signals and the simulated model output were computed for each state: position, current, and velocity. The variances of these residuals were then used to populate the diagonal elements of the Q matrix, reflecting the estimated uncertainty associated with each state. This approach allows for a data-driven tuning of the filter, directly informed by the system's real behavior.

$$Q = \begin{bmatrix} 0.00256 & 0 & 0 \\ 0 & 0.054 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The measurement noise covariance $R \in \mathbb{R}^{p \times p}$ was determined based on the resolution of the Analog-to-Digital Converter (ADC) used in the acquisition system. According to the system's datasheet specifications [2], R was computed as:

$$R = \frac{(V_{\max} - V_{\min})^2}{12 \cdot 4^N} = 1.9868e-6$$

where $V_{\max} = +10V$, $V_{\min} = -10V$, and N is the ADC resolution in bits. This expression stems from the uniform quantization error model, assuming a white, uniformly distributed quantization noise.

Kalman Gain Computation and Observer Implementation

To compute the Kalman gain, an equivalent algebraic Riccati equation formulation was used via the Linear Quadratic Regulator (LQR) duality. The observer gain was computed as:

$$L_{KF} = \text{lqr}((A + \alpha I)^T, C^T, Q, R)^T$$

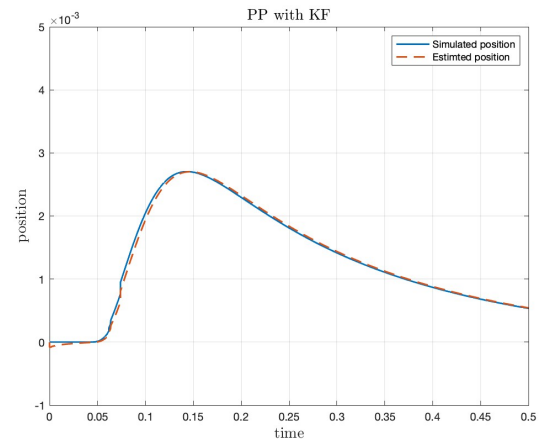
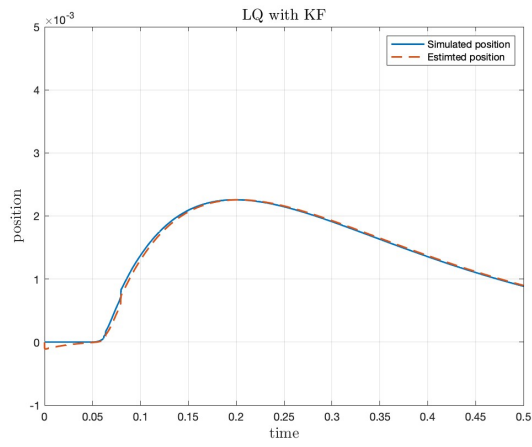
where α is a positive constant added to shift the poles for improved numerical conditioning. The poles of the observer $A - L_{KF}C$ were verified to lie in the left-half complex plane and to be significantly faster than the closed-loop poles, ensuring rapid convergence of the estimation error. The Kalman observer was then implemented in the standard observer form, extended with measurement correction terms:

$$A_{ob} = A - L_{KF}C, \quad B_{ob} = [B - L_{KF}D \quad L_{KF}], \quad C_{ob} = I, \quad D_{ob} = 0$$

This formulation allows direct simulation of the observer as a stand-alone dynamic system, facilitating integration with various control strategies and comparison with other observers.

Comparison of Kalman Filter Integration with LQ and Pole Placement Controllers

By comparing the Kalman Filter combined with LQ control and the Kalman Filter combined with pole placement in simulation, it can be observed that both approaches yield similar results. However, the more natural choice is to adopt the LQ control with the Kalman Filter because both are derived from the solution of an algebraic Riccati equation, one for estimation and the other for control. Specifically, the Kalman Filter minimizes the expected value of the estimation error covariance, while the LQ controller minimizes a quadratic cost function balancing control effort and state deviation.



Time-Domain Validation for LQ control + Kalman Filter

A step input of 4mm was applied to the system to check the performance based on real input signal.

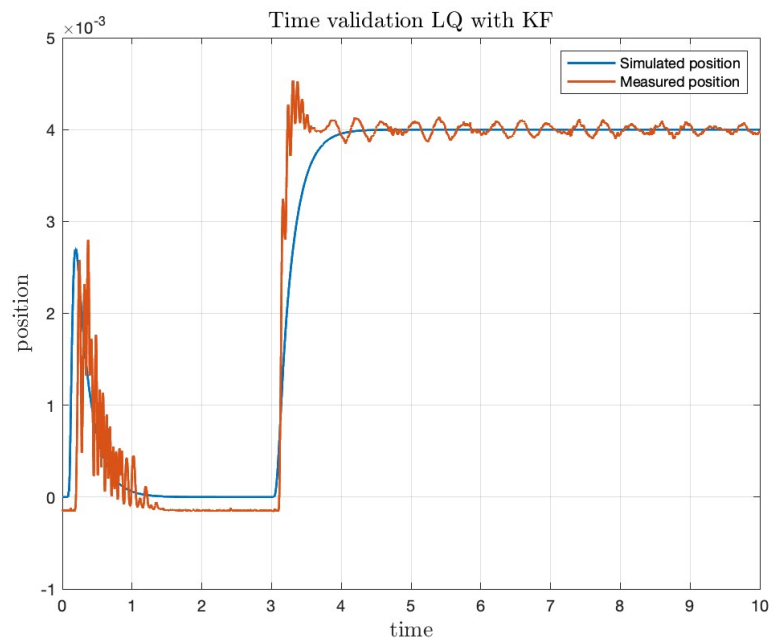


Figure 29: LQ with KF Time Validation

Frequency-Domain Validation

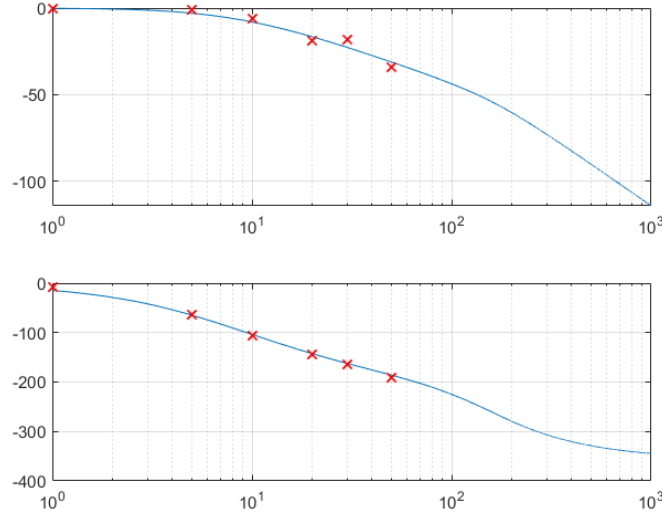


Figure 30: frequency Validation of LQ with KF

Comparison between Pole Placement with Luenberger Observer and LQ with Kalman Filter

A comparative analysis was conducted between two state-space control architectures applied to the linearized magnetic levitation system: Pole Placement with Luenberger Observer (PP+Luenberger) and Linear Quadratic Regulator with Kalman Filter (LQ+KF). The objective was to evaluate their time-domain performance using step input tests.

Linear Quadratic Regulator with Kalman Filter. This approach demonstrated noticeable oscillations and a longer settling time. The measured position exhibits a significant overshoot and sustained oscillations before reaching the steady-state. The Kalman Filter, designed using experimentally derived process noise covariance Q and measurement noise covariance R , was able to track the state, with evident noise robustness due to the stochastic nature of the Kalman Filter observer.

Pole Placement with Luenberger. The PP + Luenberger configuration showed significantly improved performance. The system response was characterized by a fast rise time, minimal overshoot, and oscillatory behavior but worst respect to the LQ+KF solution. The deterministic nature of the Luenberger to estimate the state enabled superior filtering capabilities compared to the KF observer.

Conclusion. The combination of PP control with a Luenberger is practically more consistent. This configuration proved to be more robust against disturbances, less sensitive to sensor noise, and provided better transient and steady-state performance in simulations. Therefore, the PP+Luenberger architecture is preferable for real-time control of the magnetic levitation system due to its superior reliability and overall effectiveness.

6 Task 3: Ball Position trajectory and Feedback Linearization (AC)

Trajectory Execution with Pole Placement and Luenberger Observer

The trajectory was executed using a state-space Pole Placement controller combined with a Luenberger observer. The sphere was commanded to initially perform a step from the equilibrium point to a position of 4 mm. After a few seconds, a new bias of 6 mm was imposed, upon which a sinusoidal trajectory of amplitude 1 mm and frequency 30 rad/s was superimposed. Once the sinusoidal excitation ended, the sphere returned to a reference position of 5 mm, where it was subjected to two consecutive step inputs: one descending to 4 mm and another returning to 5 mm. Finally, the sphere was excited with a sinusoidal signal of 0.5 mm amplitude and 30 rad/s frequency centered at a 4 mm bias. Despite the presence of measurement noise affecting the sphere, the transient responses were completed quickly and robustly, indicating the effectiveness of the controller-observer pair in tracking the reference under dynamic conditions.

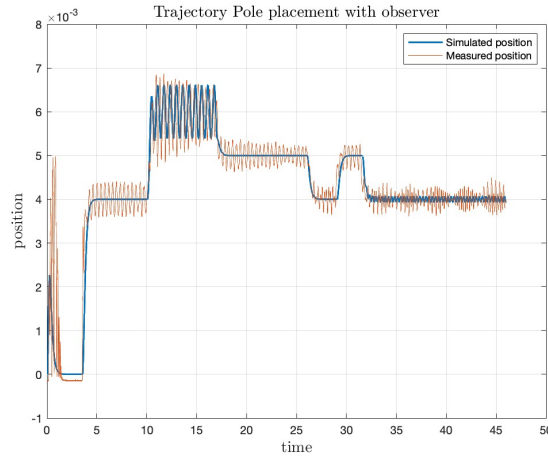


Figure 31: Trajectory executed with PP+Luenberger

6.1 Feedback Linearization

To enhance the controller's performance, an advanced technique known as *Feedback Linearization* is employed.

The main objective of this approach is to design a control law that compensates for system nonlinearities via feedback. To achieve this, a virtual control input v is defined as:

$$v = \frac{d^r y}{dt^r} \quad (12)$$

where r is the minimum relative degree for which a linear relationship between the control input u and the output y can be established, expressed as:

$$v(x) = \alpha(x) + \beta(x)u \quad (13)$$

Considering the system dynamics:

$$\begin{cases} \dot{x}_1 = x_3 \\ \dot{x}_2 = -\frac{R}{L}x_2 + \frac{u}{L} \\ \dot{x}_3 = \frac{k_m}{m} \left(\frac{x_2}{y_0 - x_1} \right)^2 - g \\ y = x_1 \end{cases}$$

The output y is differentiated until the control input u explicitly appears:

$$\dot{y} = \dot{x}_1 = x_3 \quad (14)$$

$$\ddot{y} = \dot{x}_3 = \frac{k_m}{m} \left(\frac{x_2}{y_0 - x_1} \right)^2 - g \quad (15)$$

$$\ddot{y} = \underbrace{\frac{k_m R x_2^2}{m L (y_0 - x_1)^2} + \frac{2 k_m x_2^2 x_3}{m (y_0 - x_1)^3}}_{=\alpha(x,\theta)} + \underbrace{\frac{2 k_m x_2}{m L (y_0 - x_1)^2} u}_{=\beta(x,\theta)} \quad (16)$$

At this point, input-output linearization is achieved. It is possible to cancel the system nonlinearities by setting u such that $v = \ddot{y}$:

$$u = \frac{v - \alpha(x, \theta)}{\beta(x, \theta)} \quad (17)$$

Consequently, the nonlinear system is transformed into an equivalent linear one. The controller can then be designed for the linearized system, now described by $\ddot{y} = v$.

The specific form of v depends on the chosen linear control strategy. In this work, a pole-placement approach is used:

$$\ddot{y} = \ddot{y}_{\text{ref}} - k_1 e - k_2 \dot{e} - k_3 \ddot{e} \quad (18)$$

where $e = x_1 - x_{1\text{ref}}$. The corresponding error dynamics are:

$$\ddot{e} + k_3 \dot{e} + k_2 e + k_1 e = 0 \quad (19)$$

The poles are placed at -40 , -50 , and -70 which are the same poles of the Pole Placement Controller defined in Equation 11. The gains are set to $k_1 = 140000$, $k_2 = 8300$, and $k_3 = 160$.

6.2 Steady-State Error

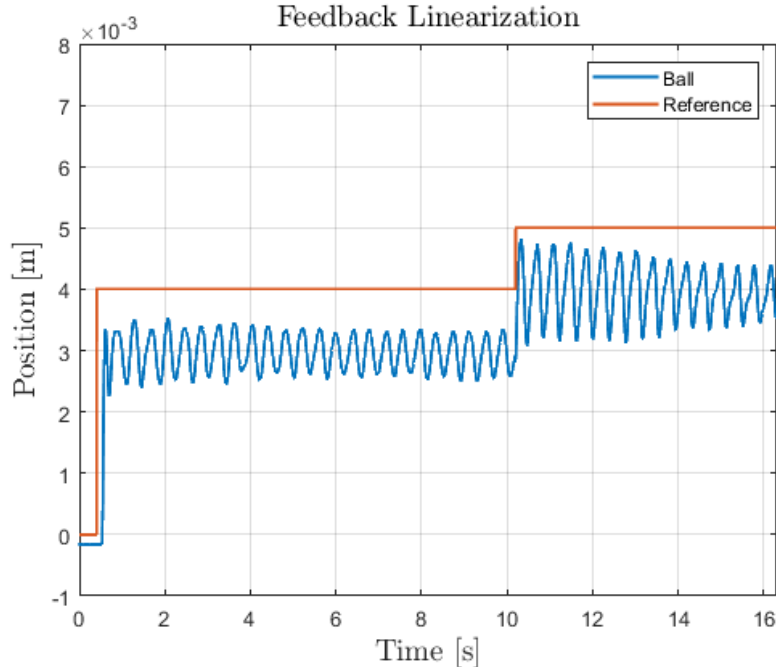


Figure 32: Step response using Feedback Linearization at 4 mm and 5 mm

In its standard form, feedback linearization does not guarantee zero steady-state error. This limitation arises because the cancellation in Equation 17 assumes a perfectly known model. However, model uncertainties and parameter variations are inevitable in practice, leading to steady-state errors as observed in Figure 32.

To mitigate this issue, two strategies are proposed: 1. Online estimation of critical parameters such as R , which may vary (e.g., due to coil heating during operation). 2. Incorporation of an integrator in the control loop to eliminate steady-state error despite model inaccuracies.

6.2.1 Adaptive Scheme

To estimate parameters online, the Recursive Least Squares (RLS) method is employed to avoid computational burdens. The relationship between R and the current dynamics is:

$$\dot{x}_2 = -\frac{R_{\text{tot}}}{L_c}x_2 + \frac{u}{L_c} \quad (20)$$

Discretizing using Euler's method gives:

$$\frac{x_2(k) - x_2(k-1)}{T_s} - \frac{u(k-1)}{L_c} = -\frac{R_{\text{tot}}}{L_c}x_2(k-1) \quad (21)$$

The regressor and parameter are defined as:

$$\phi(k-1) = -\frac{x_2(k-1)}{L_c}, \quad \theta = R_{\text{tot}}$$

So the predictor becomes:

$$\hat{y}(k) = \phi(k-1)^T \theta \quad (22)$$

In general, the least squares problem is formulated as:

$$\min_{\theta} \frac{1}{N} \sum_{k=1}^N (\tilde{y}(k) - \phi(k-1)\theta)^2 + (\theta - \theta_0)^T S (\theta - \theta_0) \quad (23)$$

where S is a regularization matrix that encodes confidence in the initial estimate θ_0 , and ensures numerical stability by preventing singularities.

To efficiently implement this online, the recursive form is used:

$$\left\{ \begin{array}{l} V(k+1) = \frac{V(k)}{\mu} - \frac{\left(\frac{V(k)}{\mu}\phi(k)\right)\left(\phi(k)^T \frac{V(k)}{\mu}\right)}{1 + \phi(k)^T \frac{V(k)}{\mu} \phi(k)} \\ \hat{\theta}(k+1) = \hat{\theta}(k) + V(k+1)\left(y(k+1) - \phi(k)^T \hat{\theta}(k)\right) \\ V_0 = S_0^{-1} = S(0)^{-1} \\ \theta_0 = \theta(0) \end{array} \right. \quad \begin{array}{l} (24) \\ (25) \\ (26) \\ (27) \end{array}$$

Here, V determines the adaptation gain in Equation 25, and is updated via Equation 24. The parameter μ is the *forgetting factor*.

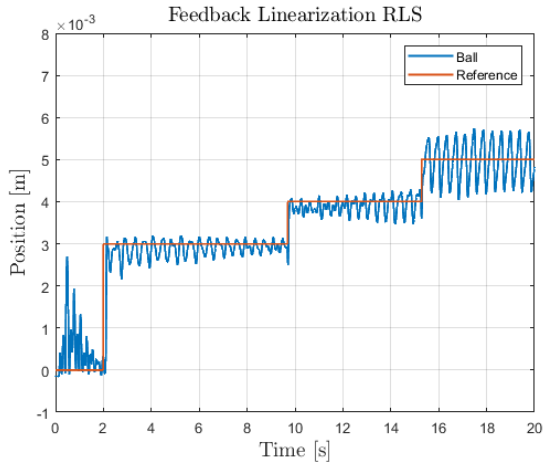
The design parameters are set as follows: θ_0 initialized from the nominal model, $S=0.1$, and $\mu=0.95$. As seen in Figure 33b, there is a mismatch between the RLS-estimated R and the nominal value used in the model. Nonetheless, the improved estimate helps significantly reduce steady-state error, as shown in Figure 33a.

6.2.2 Integral Action

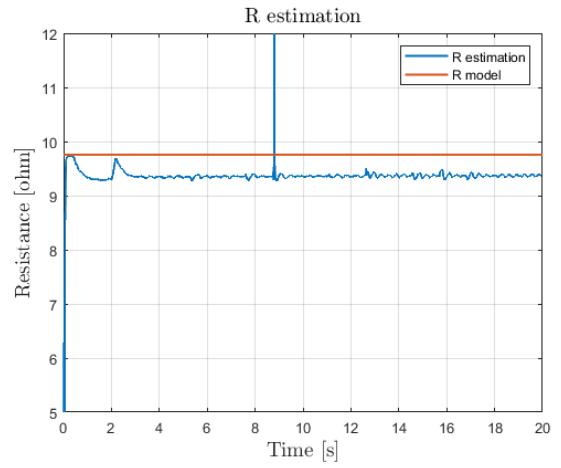
Although the adaptive scheme often eliminates steady-state error, it does not guarantee zero error under all conditions due to other uncertainties (e.g., in k_{mag} or L_c). For this reason, integral action is added to the control law Equation 19, resulting in:

$$\ddot{e} + k_3 \dot{e} + k_2 e + k_1 e + k_I \int e dt = 0 \quad (28)$$

where $k_I = 3 \times 10^5$. Since the feedback linearization already stabilizes the system, the integral action is tuned for slow convergence and low overshoot. As shown in Figure 34, the system converges in approximately 5 seconds.



(a) Feedback Linearization with RLS-based coil resistance estimation



(b) Online estimation of R during operation

Figure 33: Adaptive Feedback Linearization

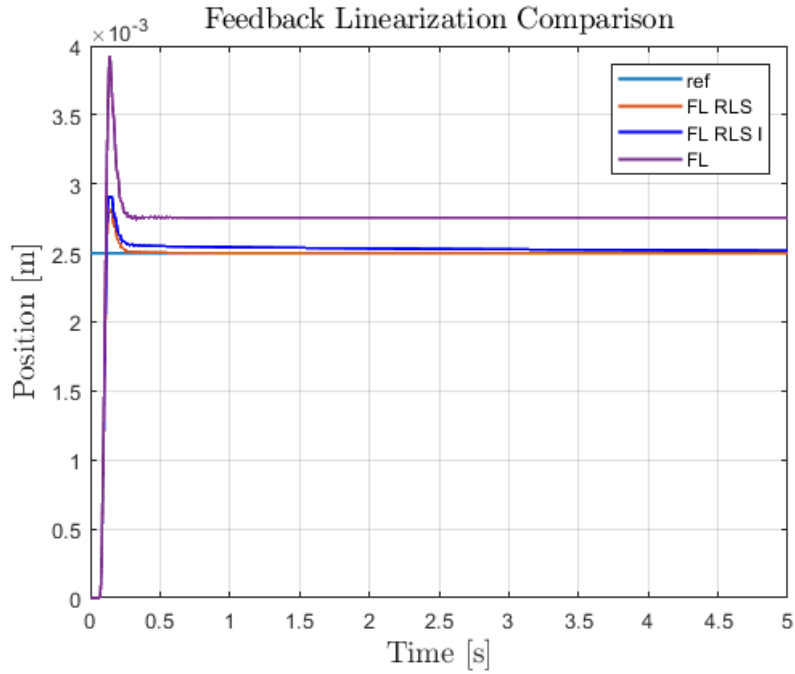


Figure 34: Comparison of Feedback Linearization Strategies

FL: Standard Feedback Linearization

FL RLS: Feedback Linearization with RLS

FL RLS I: Feedback Linearization with RLS and Integral Action

6.2.3 Results

The combined use of the adaptive scheme and integral action significantly improves performance. As shown in Figure 35, integral action is crucial in the initial phase to eliminate steady-state error. Later, the improved parameter estimation contributes to a faster response with reduced undershoot and low oscillation.

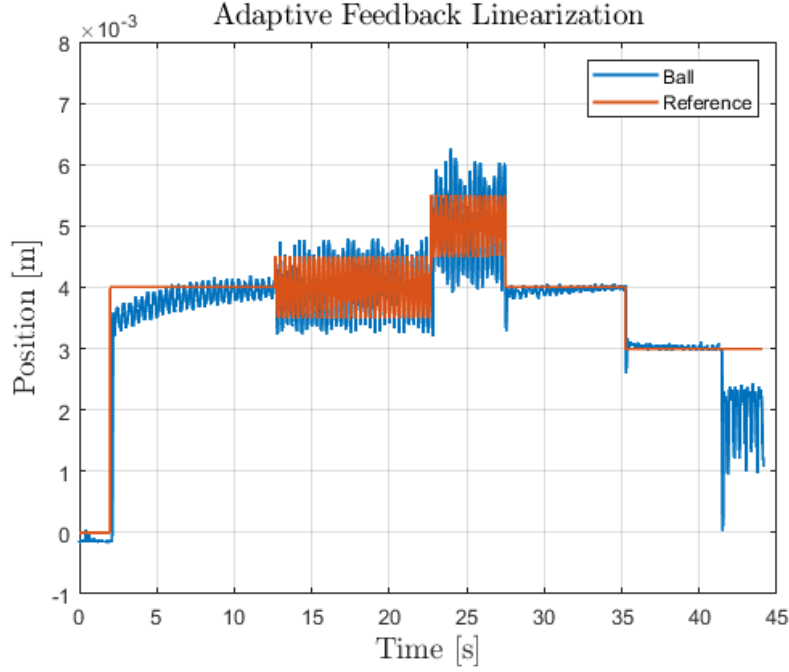


Figure 35: Final performance with Adaptive Feedback Linearization and Integral Action

Final Comparison of the Designed Controllers

At the conclusion of the laboratory activity, a critical comparison of the three main control strategies implemented can be drawn: the cascade controller (FB), the state-space controller using *pole placement* with a Luenberger observer (SS), and the *feedback linearization* controller (FL). The cascade controller demonstrated good performance, with fast settling times (< 1 s) and zero steady-state error, thanks to the clear temporal separation between the current loop and the position loop. However, it exhibits a significant overshoot, as shown in Figure 16 compared to the other strategies. The state-space controller based on *pole placement* with a Luenberger observer proved to be the most balanced solution. As illustrated in Figure 25, it satisfies all the design specifications (zero steady-state error, fast and well-damped response) and ensures excellent trajectory tracking capabilities (Figure 29). Additionally, the use of the observer enables effective estimation of unmeasured states like the velocity, resulting in a practical and high-performing implementation. The *feedback linearization* method stands out for its ability to fully linearize the nonlinear system dynamics, yielding a highly responsive behavior (Figure 30). Its effectiveness, however, depends strongly on the accuracy of the model parameters. To address this, an adaptive control scheme was implemented using a Recursive Least Squares (RLS) estimator to continuously update the solenoid's resistance in real time. This adaptation is particularly beneficial because the solenoid tends to heat up during prolonged operation, causing the resistance to vary with temperature. By incorporating this adaptive estimation, the controller can compensate for thermal variations, resulting in significantly improved robustness and responsiveness under dynamic and uncertain operating conditions. As shown in Figure 34, the incorporation of adaptive schemes and integral action enhances robustness, albeit at the cost of increased implementation complexity. In summary, the state-space controller with *pole placement* and a Luenberger observer represents the best compromise between performance, robustness, and implementation simplicity, making it the most suitable choice for real-time control of the magnetic levitation system.

References

- [1] Quanser. *USER MANUAL Magnetic Levitation Experiment*. 2012.
- [2] Quanser. *USER MANUAL Q2-USB Data Acquisition Board*. 2012.
- [3] Quanser. *USER MANUAL VoltPAQ-X1 Amplifier*. 2012.